INYANDIKO Y'UMUSHINGA · PROJECT SPECIFICATION

# Urubuga rwo gucururizaho imyambaro, inkweto n'ibyo kwambara

Isoko ry'abacuruzi benshi ryo kuri interineti ry'u Rwanda: imiterere, imikorere, ikoranabuhanga n'uko tuzubaka mu byiciro.

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Iyi nyandiko ni yo shingiro ry'ubwubatsi: uyisomye wese ashobora gutangira kubaka nta bisobanuro bindi akeneye.
Yanditswe mu Kinyarwanda; amagambo ya tekiniki n'amazina y'imbonerahamwe asigaye mu Cyongereza.
Ibyiciro: P1 (ngombwa mbere yo gufungura) · P2 · P3 · P4 (nyuma).
Igice cya 11 (amategeko) si inama y'amategeko: umunyamategeko w'u Rwanda agomba kugisoma mbere yo gufungura.

# Uko gusoma iyi nyandiko

Iyi nyandiko isobanura **umushinga wose**: icyo urubuga ari cyo, abarukoresha, buri paji n'ikiyigize, amakuru abikwa, ikoranabuhanga rikoreshwa, amategeko agomba kubahirizwa, uko tuzubaka mu byiciro, n'uko tuzamenya ko byarangiye neza. Uwayisoma wese, atigeze avugana na nyir'umushinga, agomba kuba ashobora gutangira kubaka.

**Ururimi.** Inyandiko yanditse mu Kinyarwanda. Amagambo ya tekiniki (urugero *database*, *cart*, *API*) n'amazina y'amakuru n'ay'imbonerahamwe (urugero `products`) asigaye mu Cyongereza, kuko ari ko abubaka bayamenyereye kandi ari ko azandikwa muri code.

**Uko inyandiko iteguwe.**

- **Ibice 1 kugeza kuri 4** bisobanura *igitekerezo*: icyo dukora, impamvu, abakoresha n'ingendo zabo.
- **Ibice 5 kugeza kuri 7** bisobanura *icyo tubaka*: ibisabwa byose, ifishi yo gushyiraho ibicuruzwa, n'amakuru abikwa.
- **Ibice 8 kugeza kuri 10** bisobanura *uko tubaka*: ikoranabuhanga, algorithm, kugenzura abacuruzi n'umutekano.
- **Ibice 11 kugeza kuri 13** bisobanura *amategeko, isura y'urubuga, n'uko tuzubaka mu byiciro*.
- **Ibice 14 kugeza kuri 17** bisobanura *ibizamini, ibyago, amafaranga, n'ibibazo bigomba kwemezwa*.
- **Ibindi bifatanye (Ibyongeweho A kugeza kuri D)** birimo inyandiko z'ubutumwa bwa WhatsApp, urutonde rw'uturere n'imirenge, amabwiriza y'amafoto, n'aho amakuru yavuye.

> **Ikimenyetso cy'ibyiciro**
> Ahantu hose hari ikimenyetso cy'icyiciro: **P1** = ngombwa mbere yo gufungura (icyiciro cya 1); **P2** = icyiciro cya 2 (gufungura abacuruzi benshi); **P3** = icyiciro cya 3 (kuzamura); **P4** = nyuma yaho. Ibyo mu P1 ni byo byonyine bigomba kubakwa mbere. Ibindi ntibigomba gutangira mbere y'uko P1 irangira, ariko imiterere y'amakuru n'ubwubatsi byateguwe kugira ngo bizongerwemo nta gutangira bundi bushya.

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# 1. Incamake y'umushinga

## 1.1 Icyo urubuga ari cyo

Urubuga ni **isoko ryo kuri interineti ry'u Rwanda**, ryihariye imyambaro, inkweto, amasogisi n'ibindi byo kwambara (nyuma yaho: ibikoresho by'amatelefoni). Ni 'ikibuga' abacuruzi baza gucururizaho. Umucuruzi wemewe ashyiraho amafoto, ibiciro n'imibare (sizes) y'ibicuruzwa bye. Umukiriya wese arareba, agatoranya, agashyira mu gare, hanyuma akavugana n'umucuruzi kuri WhatsApp cyangwa kuri telefoni.

**Kwishyura, kumvikana ku giciro no gutanga ibicuruzwa bikorwa hagati y'umukiriya n'umucuruzi.** Urubuga ntirwinjira muri ayo masezerano, ntirwakira amafaranga y'abandi kandi nta komisiyo rwishyuza.

Urubuga rutangirana n'abacuruzi batatu: nyir'umushinga n'abacuruzi babiri b'inshuti zizewe, i Kigali. Rwubakwa kugira ngo rushobore gukura rukaba isoko ry'igihugu cyose.

## 1.2 Ikibazo gikemurwa

- Abacuruzi bashyira ibicuruzwa kuri **status za WhatsApp**. Status ibura nyuma y'amasaha 24, nta rutonde ruhoraho, nta gushakisha, nta kugereranya.
- Buri munsi bakubazwa kimwe: '*Angahe?*', '*Size zihari?*', '*Irahari?*'. Ibi bibatwara umwanya kandi bituma batakaza abakiriya batitabwaho vuba.
- Umukiriya ntabona ahantu hamwe ibicuruzwa by'abacuruzi benshi, kandi ntiyizera byoroshye umucuruzi atazi.

**Igisubizo:** ububiko buhoraho bw'ibicuruzwa bufite amafoto, ibiciro, imibare n'icyo gihari/gishize; link imwe y'iduka rya buri mucuruzi; igare rigabanya ibintu ku mucuruzi; n'abacuruzi bagenzuwe.

## 1.3 Ibyemezo by'ingenzi byafashwe

| # | Ingingo | Icyemezo |
|---|---|---|
| 1 | Ubwoko bw'urubuga | Isoko ry'abacuruzi benshi (marketplace) ryo kuri interineti. Gitangirana n'abacuruzi 3. |
| 2 | Uko igurisha rikorwa | Umukiriya avugana n'umucuruzi kuri WhatsApp cyangwa kuri telefoni. **Nta kwishyura ku rubuga.** |
| 3 | Komisiyo | Nta komisiyo. Urubuga ntirwinjira mu bucuruzi bw'abandi. |
| 4 | Amafaranga yo gukoresha urubuga | Mu ntangiriro ni ubuntu kuri bose. Nyuma y'imyaka (3 kugeza kuri 5) urubuga rumaze kumenyekana: amatangazo (nka Google AdSense), hanyuma ubukode bw'ukwezi bw'abacuruzi (reba igice 16). |
| 5 | Delivery | Ni umucuruzi n'umukiriya bayumvikanaho. Urubuga ntirwizeza igihe cyo gutanga. |
| 6 | Abakiriya | **Nta konti.** Igare, ibyo basuye n'aho bari bibikwa kuri telefoni yabo. |
| 7 | Abacuruzi | Bafungura konti kandi bakagenzurwa (indangamuntu, telefoni, ifoto y'aho bacururiza, ibyo bacuruza) mbere y'uko ibicuruzwa byabo bigaragara. |
| 8 | Ahantu | Kigali mbere (P1, P2). Igihugu cyose nyuma (P3). |
| 9 | Ururimi | Kinyarwanda mbere. Icyongereza gishyirwamo mu cyiciro cya 3. |
| 10 | Ibikoresho bya tekiniki | Serivisi z'ubuntu mu ntangiriro. Gukoreshwa kuri telefoni mbere. Nta app yo gukuramo: PWA (reba igice 8). |
| 11 | Imiterere y'amakuru | Yubakwa ngo abacuruzi benshi bashobore kongerwamo guhera ku munsi wa mbere, kugira ngo tutazatangira bundi bushya. |

## 1.4 Ibyo urubuga rutazakora

Ibi byanditswe kugira ngo uwubaka atabyubaka kubera ko 'bisanzwe mu bindi bibuga'. Ntibigomba kubakwa mu byiciro P1, P2 na P3 keretse nyir'umushinga abyemeje mu nyandiko.

- **Kwishyura ku rubuga** (MoMo, amakarita, wallet) n'**igare ryishyurwa**. Igare ni urutonde rwo gutoranya gusa.
- **Kubika amafaranga y'abandi** no kuyatanga igicuruzwa kigeze (escrow).
- **Delivery** y'urubuga: nta batwara b'urubuga, nta gukurikirana urugendo.
- **Chat y'imbere mu rubuga.** Ibiganiro byose bijya kuri WhatsApp.
- **Konti z'abakiriya** n'amagambo y'ibanga y'abakiriya.
- **App ya Android cyangwa iOS** mu maduka ya Google/Apple. PWA irahagije.
- **Ibicuruzwa bitari ibyo kwambara cyangwa ibikoresho by'amatelefoni** (ibiribwa, imodoka, amazu...). Icyiciro 'Ibikoresho by'amatelefoni' ni P3.
- **Gucuruza hanze y'u Rwanda.**

## 1.5 Amagambo akoreshwa

| Ijambo | Icyo risobanura |
|---|---|
| **Urubuga** | Iyi platform yose: ahagaragara ku bakiriya, ahacungirwa n'abacuruzi, n'ahacungirwa na nyir'urubuga. |
| **Umucuruzi (seller)** | Umuntu cyangwa iduka ryafunguye konti rigashyiraho ibicuruzwa. |
| **Iduka (shop)** | Urupapuro rw'umucuruzi rufite izina, ifoto, aho acururiza, nimero ya WhatsApp n'ibicuruzwa bye. Mu cyiciro cya 1 umucuruzi 1 afite iduka 1. |
| **Igicuruzwa (product)** | Ikintu kigurishwa (urugero: inkweto z'ubwoko bumwe), gifite amafoto, igiciro n'ibyiciro. |
| **Variant** | Ubwoko bwihariye bw'igicuruzwa: size + ibara (urugero: size 42, umukara). |
| **Igare (cart)** | Urutonde rw'ibyo umukiriya yatoranyije, rubikwa kuri telefoni ye. Ntirwishyurwa. |
| **Umukiriya (buyer)** | Umuntu wese usura urubuga. Nta konti agira. |
| **Admin / Nyir'urubuga** | Ugenzura abacuruzi, akemura ibirego, akayobora urubuga. |
| **KYC** | Kugenzura umucuruzi: amazina, indangamuntu, telefoni, ifoto y'aho acururiza. |
| **PWA** | Urubuga rushyirwa kuri home screen ya telefoni rugakora nka app, rugakoresha data nke. |
| **Link ya wa.me** | Link ifungura WhatsApp ifite ubutumwa bwamaze kwandikwa. Ni ubuntu. |
| **RLS** | Row Level Security: amategeko ya database yemeza ko buri wese abona gusa amakuru yemerewe. |
| **Push notification** | Ubutumwa buza kuri telefoni umukiriya yemeye kubwakira, na iyo urubuga rudafunguye. |
| **RWF** | Amafaranga y'u Rwanda. Ibiciro byose bibikwa nk'imibare yuzuye (integer), nta mbibi. |
| **MVP** | Verisiyo ya mbere ntoya ikora neza ikoreshwa n'abacuruzi 3 (icyiciro cya 1). |

# 2. Intego n'amahame y'imikorere

## 2.1 Intego

**Intego nkuru:** gukora urubuga abantu benshi biyumvamo, rworoheje, rwihuta, rutagira amananiza, kugira ngo rukurure abantu. Amafaranga aza nyuma y'uko urubuga rumaze kumenyekana.

| Igipimo | Intego y'icyiciro cya 1 (nyuma y'ibyumweru 4 rufunguye) | Uko tukigenzura |
|---|---|---|
| Igihe cyo gushyiraho igicuruzwa | Amasegonda 60 cyangwa munsi, kuva ku ifoto kugeza ku 'Bika' | Dupima n'abacuruzi 3, kuri telefoni zabo |
| Igihe urupapuro rw'imbere rufungukira | Munsi y'amasegonda 3 kuri telefoni isanzwe ya Android kuri 4G itinze; nyuma y'aho munsi y'isegonda 1 (kuva muri cache) | Chrome DevTools (throttling) na telefoni nyayo |
| Ibicuruzwa biriho | Nibura 60 | Admin dashboard |
| Abasura | Nibura 300 mu cyumweru cya mbere (biva ku status z'abacuruzi 3 n'inshuti) | Cloudflare Web Analytics |
| Abakanze kuri WhatsApp | Nibura 50 mu cyumweru cya mbere | shop_daily_stats, product_daily_stats |
| Ibirego bitarakemurwa mu masaha 24 | 0 | Admin dashboard |

*Izi ni intego z'ibanze zo kuyobora, si amasezerano. Zihindurwa nyuma y'ibyumweru 2 bya mbere.*

## 2.2 Amahame y'imikorere

1. **Telefoni mbere.** Abakiriya n'abacuruzi benshi bakoresha telefoni za Android zisanzwe. Buri paji igomba gukora neza ku cyerekezo cya 360 px z'ubugari. Computer ni iy'ingoboka.
2. **Data nke.** Amafoto yose agabanywa mbere yo koherezwa kandi akerekanwa mu bunini bukwiye (reba 8.6). Urubuga rwose rwa mbere rugomba kuba munsi ya 500 KB.
3. **Kinyarwanda mbere.** Amagambo yose y'ahagaragara ni Kinyarwanda cyoroshye, kitarimo amagambo y'ikoranabuhanga.
4. **Nta mananiza.** Umukiriya ntakora konti kugira ngo ahahe. Umucuruzi atanga amakuru gusa akenewe.
5. **Icyizere.** Umucuruzi wese agaragara afite ikimenyetso cya 'Yagenzuwe' nyuma yo kugenzurwa. Igicuruzwa cyose gifite buto yo gutanga ikibazo.
6. **Ukuri.** Ibiciro, ibisigaye n'imiterere (nshya cyangwa mitumba) bigomba kuba ari ukuri. Nta bwoko bw'ibinyoma bwemewe (reba igice 10).
7. **Kuramba ku buntu.** Dukoresha serivisi z'ubuntu kandi tukagenzura imipaka yazo (reba 8.4). Nta cyemezo cya tekiniki gifata uruhande ruhenze mu cyiciro cya 1.
8. **Kubaka gato, gukura buhoro.** Buri cyiciro kigerageza ikintu kimwe kitaragerwaho (abacuruzi bazaza? abakiriya bazagura? bazagaruka?).

## 2.3 Abakoresha n'ibyo bakeneye

| Uwo ari we | Imiterere | Icyo akeneye ku rubuga |
|---|---|---|
| **Umukiriya** (urugero: umukobwa w'imyaka 24 i Kimironko) | Telefoni ya Android ifite 1 GB y'ibyo agura buri cyumweru. Akoresha WhatsApp buri munsi. Atizera umucuruzi atazi. | Kubona vuba ibimubereye (size ye), igiciro kigaragara, ifoto nziza, kumenya uwo agurira, n'ukumuhamagara cyangwa kumwandikira byoroshye. |

| Uwo ari we | Imiterere | Icyo akeneye ku rubuga |
|---|---|---|
| **Umucuruzi** (urugero: ucuruza inkweto aturutse mu rugo cyangwa mu iduka rito) | Akoresha telefoni gusa. Ashyira ibicuruzwa kuri status ya WhatsApp na Instagram/Facebook. Abonye abantu benshi bamubaza ibiciro. | Gushyiraho igicuruzwa mu gihe gito, link imwe y'iduka yo gusangiza, kubona uwarebye ibye n'uwakanze WhatsApp, no kuzimya igicuruzwa cyashize byoroshye. |
| **Nyir'urubuga** | Ni we ufite ubushobozi bwose. Mu ntangiriro ni umuntu umwe. | Kwemeza abacuruzi, guhisha ibicuruzwa bibi, gukemura ibirego, kubona imibare y'urubuga, no kwirinda gukora byinshi mu ntoki. |

# 3. Uruhare n'uburenganzira

Hari uruhare enye (roles). Uburenganzira bwose bushyirwa mu bikorwa mu rwego rwa database (RLS, reba 7.3) si mu mpapuro z'urubuga gusa, kugira ngo umuntu atabushobora gukoresha uburyo bwo kunyura inzira ndende.

| Igikorwa | Umukiriya (nta konti) | Umucuruzi (utaremezwa) | Umucuruzi (wemejwe) | Admin |
|---|---|---|---|---|
| Kureba ibicuruzwa n'amaduka byemejwe | Yego | Yego | Yego | Yego |
| Gukoresha igare, gushakisha, aho ari | Yego | Yego | Yego | Yego |
| Gukanda WhatsApp / guhamagara | Yego | Yego | Yego | Yego |
| Gutanga ikibazo (report) | Yego | Yego | Yego | Yego |
| Kwiyandikisha nk'umucuruzi no kohereza ibyangombwa | — | Yego | — | — |
| Gushyiraho ibicuruzwa | — | Yego, ariko ntibigaragara | Yego, bigaragara ako kanya | Yego (ku bwa se) |
| Guhindura no gusiba ibicuruzwa bye | — | Ibye gusa | Ibye gusa | Byose |
| Kureba imibare ye | — | — | Yego (ye gusa) | Yose |
| Kureba indangamuntu n'ibyangombwa | — | Ibye gusa | Ibye gusa | Yego (byandikwa mu audit_log) |
| Kwemeza, guhagarika, guhisha umucuruzi | — | — | — | Yego |
| Guhisha igicuruzwa icyo ari cyo cyose | — | — | — | Yego |
| Kureba no gukemura ibirego | — | — | — | Yego |
| Guhindura ibyiciro, uturere, amategeko | — | — | — | Yego |
| Kohereza notification kuri bose | — | — | — | Yego |

**Icyitonderwa ku 'mucuruzi utaremezwa'**
Mu cyiciro cya 1, abacuruzi 3 bemezwa na nyir'urubuga ubwe mbere yo gutangira, bityo iyi ntambwe ntibabarwa. Mu cyiciro cya 2, umucuruzi mushya ashobora kwandika ibicuruzwa mbere yo kwemezwa ariko bigaragara gusa nyuma yo kwemezwa (reba 10.1).

# 4. Ingendo z'abakoresha

## 4.1 Urugendo rw'umukiriya

1. **Kuza ku rubuga.** Akenshi akanze link ku status y'umucuruzi (igisubizo kimwe: ajya ku rupapuro rw'igicuruzwa cyangwa iduka). Ashobora kandi kuza ku rupapuro rw'imbere.
2. **Gushakisha.** Ahitamo icyiciro (Inkweto, Imyenda...) cyangwa agashakisha izina ('nike', 'ankara'). Ashobora kungurura ku *size*, igiciro, ibara, imiterere (nshya/mitumba) n'aho ari.
3. **Kureba igicuruzwa.** Abona amafoto (kunyerera), izina, igiciro (n'icya kera kiriho umurongo n'ijanisha), sizes zihari, ibisigaye (niba ari bike), iduka n'ikimenyetso 'Yagenzuwe', n'ibisa n'icyo.
4. **Guhitamo size n'ibara** hanyuma agakanda **'Shyira mu gare'**, cyangwa **'Vugana n'umucuruzi'** (ubutumwa bw'igicuruzwa kimwe gusa).
5. **Igare.** Igare rigabanywa ku mucuruzi. Umukiriya abona: ibyo yatoranyije, umubare, igiteranyo kuri buri mucuruzi, n'icyitonderwa 'Kwishyura no gutanga bikorwa hagati yawe n'umucuruzi'.
6. **Aho ari.** Mbere yo kohereza, asabwa guhitamo akarere (n'umurenge i Kigali) n'akarango (si itegeko). Ashobora gukanda 'Ohereza aho ndi' (GPS) si itegeko. Ibi bibikwa kuri telefoni ye gusa.
7. **Kohereza.** Kuri buri mucuruzi mu gare akanda 'Ohereza kuri WhatsApp ya [iduka]'. WhatsApp irafunguka ubutumwa bwamaze kwandikwa (reba Ibyongeweho A). Ashobora no guhamagara.
8. **Nyuma.** Umukiriya n'umucuruzi bumvikana (igiciro, delivery, kwishyura). Igare ry'uwo mucuruzi rirakurwaho (cyangwa agahitamo kurigumana).
9. **Kugaruka.** Urupapuro rw'imbere rumwereka 'Wari warabirebye' n'ibisa n'ibyo yakunze. Niba yemeye notification, ahabwa ibishya.

### Ibihe bidasanzwe umukiriya ashobora guhura na byo

| Ikibazo | Uko urubuga rukora |
|---|---|
| Internet igiye mu gihe aho ari | Ibyo yari yarebye n'igare biracyagaragara (cache). Ubutumwa: 'Nta internet. Ibyo wari warebye biracyahari.' |
| Igicuruzwa cyashize nyuma yo gishyira mu gare | Mu gare kigaragara 'Cyashize' ntikibarwe mu giteranyo, ariko ntigisibwa; aho bishoboka, hagaragara ibisa na cyo. |
| Size atoranyije yashize | Buto ya 'Shyira mu gare' ntikora kuri iyo size; size zihari gusa ni zo zikandwa. |
| Umucuruzi yahagaritswe | Ibicuruzwa bye bihita bihishwa. Mu gare hagaragara 'Iduka ntirikiriho'. |
| Nta WhatsApp muri telefoni | Link ya wa.me ifungura urubuga rwa WhatsApp Web cyangwa ikamusaba kuyishyira. Hongerwaho buto 'Hamagara' na 'Koporora ubutumwa'. |
| Link y'igicuruzwa yasibwe | Urupapuro 'Iki gicuruzwa ntikiriho' n'ibisa na cyo n'ibindi by'iduka ryacyo. |

## 4.2 Urugendo rw'umucuruzi

1. **Kwiyandikisha** (reba 10.1): mu cyiciro cya 1 nyir'urubuga afungurira konti abacuruzi 3. Mu cyiciro cya 2: umucuruzi yuzuza ifishi y'ibyangombwa.
2. **Kwinjira:** nimero ya telefoni n'ijambo ry'ibanga (nibura imibare/inyuguti 6). Telefoni imenya uwinjiye ntisabe kongera kwinjira igihe cy'iminsi 30.
3. **Gushyiraho igicuruzwa** mu ntambwe 4 (reba igice 6): ifoto → izina n'igiciro → sizes n'amabara → ugusuzuma no 'Bika'. Intego: amasegonda 60.
4. **Gucunga:** 'Kopora' (gukora igicuruzwa gisa), 'Cyashize' (buto imwe), guhindura igiciro, guhindura sizes, gusiba. Iyo atakoze ikintu ku gicuruzwa mu minsi 30, abazwa 'Iracyahari?'.

5. **Gusangiza:** kopora link y'iduka, kubona QR code y'iduka, no gusangiza igicuruzwa kuri WhatsApp status (link ifite ifoto n'igiciro, reba 8.8).
6. **Imibare:** uko ibicuruzwa bye byarebwe n'inshuro zakanzwe kuri WhatsApp mu minsi 7 na 30.
7. **Kubona ubutumwa bw'urubuga:** bugaragara ku rupapuro rw'imbere rwe (urugero 'Ibyangombwa byawe byemejwe', 'Igicuruzwa cyawe cyatanzweho ikibazo').

## 4.3 Urugendo rwa Admin

1. **Buri munsi:** kureba 'Abategereje kwemezwa', 'Ibirego bishya', 'Ibicuruzwa bishya'.
2. **Kwemeza umucuruzi:** gufungura ibyangombwa, guhuza isura iri kuri selfie n'iri ku ndangamuntu, guhamagara nimero y'umucuruzi no kwemeza ko ari iye, kureba ifoto y'aho acururiza, hanyuma 'Emeza', 'Subiza inyuma (asabwe gukosora)' cyangwa 'Anga' ushyiraho impamvu.
3. **Gukemura ikibazo:** gusoma ikirego, kureba igicuruzwa n'umucuruzi, guhisha igicuruzwa cyangwa guhagarika iduka, kwandikira uwatanze ikirego (iyo yatanze nimero), no kwandika icyakozwe.
4. **Buri cyumweru:** kureba imibare, kugenzura imipaka y'ubuntu (reba 8.4), no kureba kopi y'ububiko (backup).

# 5. Ibisabwa mu mikorere

Buri gisabwa gifite nimero. **Nimero** igizwe n'inyuguti y'igice n'umubare (urugero **C-08**). Uwubaka n'uwipima bakoresha izi nimero. **P1**, **P2**, **P3**, **P4** ni ibyiciro (reba igice 13).

## A. Urupapuro rw'imbere (Home)

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **A-01** | Umutwe uhoraho: logo, akazu ko gushakisha, akarere umukiriya arimo (gukandwa ngo ahindure), n'igare rifite umubare w'ibirimo. | P1 |
| **A-02** | Ibyiciro bikuru nk'utubuto tunini dufite ishusho: Inkweto, Imyenda, Amasogisi n'imyenda y'imbere, Ibindi byo kwambara. (Ibikoresho by'amatelefoni: P3.) | P1 |
| **A-03** | Imirongo y'ibicuruzwa itambitse (yanyerera): 'Wari warabirebye' (iyo ihari), 'Bishya', 'Bikunzwe cyane', 'Byagabanyijwe', 'Amaduka hafi yawe'. Buri mugongo ugaragaza ibicuruzwa 8 gusa n'akabuto 'Reba byose'. | P1 |
| **A-04** | Ibicuruzwa biremerwa buhoro buhoro (20 buri gihe) hamwe n'akabuto 'Reba byinshi'. Nta kuzamura kutagira iherezo (infinite scroll) mu cyiciro cya 1, kuko gatuma data itakara. | P1 |
| **A-05** | Itangazo ryo gushyira urubuga kuri home screen (PWA) rigaragara nyuma y'uko umukiriya asuye ibihe 2, kandi rishobora kwangwa burundu. | P2 |
| **A-06** | Hasi ya buri paji: 'Ushaka gucuruza?' (ijya ku mabwiriza y'abacuruzi), Ibyerekeye, Amategeko, Politiki y'ibanga, 'Ubufasha kuri WhatsApp'. | P1 |
| **A-07** | Buto 'Dukurikire kuri WhatsApp Channel' ku rupapuro rw'imbere no hasi. | P3 |

## B. Gushakisha n'ibyungurura

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **B-01** | Akazu ko gushakisha gatanga ibyifuzo igihe umukiriya yandika (ibyiciro, ibirango, amazina y'ibicuruzwa), kandi gakora na nyuma y'inyuguti 2. | P1 |
| **B-02** | Gushakisha bihanganira amakosa yo kwandika n'amagambo ahwanye (urugero 'nayiki' ibonwa nka Nike; 'isogisi' na 'amasogisi'). Reba 9.1. | P1 |
| **B-03** | Ibyungurura: icyiciro, igiciro (kuva–kugeza), size, ibara, imiterere (nshya/mitumba), akarere, 'biri mu bubiko gusa', 'byagabanyijwe gusa', iduka. | P1 |
| **B-04** | Ibyungurura bigaragara mu rupapuro rwo hasi (bottom sheet) kuri telefoni, kandi umubare w'ibyungurujwe ugaragara ku kabuto. | P1 |
| **B-05** | Gutondeka: Bishya, Igiciro kiva ku gito, Igiciro kiva ku kinini, Bikunzwe cyane, Hafi yanjye. | P1 |
| **B-06** | Ibyungurura n'ibyo gushakisha bibikwa muri URL (urugero `/shakisha?q=nike&size=42&igiciro=20000-50000`) kugira ngo bishobore gusangizwa. | P1 |
| **B-07** | Iyo nta bisubizo: ubutumwa busobanutse, ibyifuzo by'ibisa, n'akabuto ko kuvanaho ibyungurura. | P1 |
| **B-08** | 'Size yanjye': umukiriya ahitamo size y'inkweto n'iy'imyenda inshuro imwe ikibikwa kuri telefoni, ikungurura impapuro zose ubwayo (ashobora kuyizimya). | P2 |

## C. Urupapuro rw'igicuruzwa

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **C-01** | Amafoto 1 kugeza kuri 5: kunyerera ibumoso/iburyo, gukanda ngo yiyongere, kwerekana nyuma y'ifoto ya mbere (lazy). Ifoto ya mbere igaragara mu 'thumbnail' mu mirongo no mu gushakisha. | P1 |

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **C-02** | Izina, igiciro cy'ubu (RWF, nka '45 000 RWF'). Iyo hari igiciro cya kera: kigaragara gifite umurongo hejuru n'ijanisha ry'ibyagabanutse (urugero '-20%'), bibarwa n'urubuga. | P1 |
| **C-03** | Sizes zerekanwa nk'utubuto. Izihari zirakandwa; izashize zirazimye ariko zigaragara. Ntawe ushobora gukanda size yashize. | P1 |
| **C-04** | Amabara yerekanwa nk'utuzenguruko cyangwa utubuto turi mu nyandiko; guhitamo ibara bihindura ifoto aho umucuruzi yahuje ifoto n'ibara (P3). | P1 |
| **C-05** | 'Hasigaye N' (N ≤ 3) kigaragara iyo umucuruzi yanditse umubare uriho. Ntikigaragara iyo umucuruzi atanditse umubare. | P1 |
| **C-06** | Imiterere: 'Nshya' cyangwa 'Mitumba'; ikirango (iyo kiriho) n'icyo gifite (reba 6.3); ibisobanuro bitarenze inyuguti 300. | P1 |
| **C-07** | Ikarita y'umucuruzi: izina ry'iduka, ikimenyetso 'Yagenzuwe', aho rihererereye (akarere/umurenge), buto 'Reba iduka'. | P1 |
| **C-08** | Utubuto duhoraho hasi y'ecran: 'Shyira mu gare' (nyamukuru), 'WhatsApp' (ubutumwa bw'igicuruzwa kimwe), 'Hamagara'. | P1 |
| **C-09** | 'Bigurwa hamwe': ibindi by'uwo mucuruzi bijyana n'icyo (urugero amasogisi hamwe n'inkweto) n'icyifuzo 'Ongeraho kuri 2 000 RWF'. | P2 |
| **C-10** | 'Ibisa na cyo': ibicuruzwa 8 (reba 9.2). | P1 |
| **C-11** | 'Tanga ikibazo kuri iki gicuruzwa' (reba L) n'inama z'umutekano ngufi ('Reba igicuruzwa mbere yo kwishyura iyo muri Kigali'). | P1 |
| **C-12** | Gusangiza: akabuto ko gukoporora link n'ako gusangiza kuri WhatsApp. Link igaragara kuri WhatsApp ifite ifoto, izina n'igiciro (reba 8.8). | P1 |
| **C-13** | Igicuruzwa cyashize kigaragara 'Cyashize' (ifoto isa n'izimye), ntigishyirwa mu gare, kikamara iminsi 14 gisigaye aho link ziri, hanyuma kigahishwa. | P1 |
| **C-14** | Kubara 'view' imwe gusa kuri buri session kuri buri gicuruzwa ku munsi (reba M). | P1 |

## D. Urupapuro rw'iduka

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **D-01** | Umutwe w'iduka: logo, izina, aho rihererereye, ikimenyetso 'Yagenzuwe', itariki ryatangiriyeho, umubare w'ibicuruzwa biriho, inyandiko yo 'Gutanga' (urugero 'Gutanga i Kigali na hanze'). | P1 |
| **D-02** | Utubuto: 'WhatsApp', 'Hamagara', 'Sangiza iduka'. | P1 |
| **D-03** | Urutonde rw'ibicuruzwa by'iduka rufite ibyungurura bya B. | P1 |
| **D-04** | Link ngufi y'iduka: `/i/izina-ry-iduka` (slug idasubirwamo, inyuguti nto, imibare n'utuyobora gusa). | P1 |
| **D-05** | QR code y'iduka ikorerwa mu mushakisha, igakururwa nk'ifoto, kugira ngo umucuruzi ayicape ayishyire ku iduka rye. | P2 |
| **D-06** | 'Kurikira iduka': umukiriya yemera notification z'iduka runaka. | P3 |

## E. Igare

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **E-01** | Akabuto k'igare kagaragara ku mpapuro zose, gafite umubare w'ibirimo. | P1 |
| **E-02** | Igare rigabanywa ku mucuruzi: buri mucuruzi afite igice cye cyanditseho izina ry'iduka n'igiteranyo cyacyo. Nta giteranyo rusange cyemeza ko bishyurwa hamwe. | P1 |

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **E-03** | Buri murongo: ifoto nto, izina, size, ibara, igiciro, umubare (+/−), kuvanaho. | P1 |
| **E-04** | Igare ribikwa muri `localStorage` iminsi 30, rigakomeza hagati y'impapuro n'ama-tab. | P1 |
| **E-05** | Igihe igare rifunguwe, urubuga rugenzura kuri seriveri ibiciro n'ibisigaye. Ibyahindutse bigaragazwa ('Igiciro cyahindutse: 40 000 → 38 000'); ibyashize bigaragazwa 'Cyashize' ntibibarwe. | P1 |
| **E-06** | Ku mucuruzi wese: buto 'Ohereza kuri WhatsApp ya [iduka]' ifungura `wa.me` n'ubutumwa bwanditswe (Ibyongeweho A). Buto 'Hamagara' na 'Koporora ubutumwa' ziriho. | P1 |
| **E-07** | Icyitonderwa gihoraho mu gare: 'Kwishyura no gutanga bikorwa hagati yawe n'umucuruzi. Urubuga ntiruhagarikira kwishyura.' | P1 |
| **E-08** | Imipaka: ibintu bitarenga 20 mu gare, 10 ku mucuruzi (kugira ngo ubutumwa bwa WhatsApp budasagarira). | P1 |
| **E-09** | Buri gukanda WhatsApp/Hamagara kwandikwa nk'ikintu cyo kubara (reba M), nta makuru bwite. | P1 |
| **E-10** | Nyuma yo kohereza, ibintu by'uwo mucuruzi bishobora gukurwaho ('Kuraho ibyoherejwe') cyangwa bikagumaho. | P2 |

## F. Aho umukiriya ari

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **F-01** | Urupapuro ruto rusaba: Akarere (kuri 30), Umurenge (i Kigali mu cyiciro cya 1; ahandi P3), Akarango (inyandiko, ≤80 inyuguti, si itegeko). Rugaragara bwa mbere asaba kohereza kuri WhatsApp no kuva ku rupapuro rw'imbere ('Hahe uri?'). | P1 |
| **F-02** | 'Ohereza aho ndi' ikoresha GPS ya mushakisha igakora link ya Google Maps. Si itegeko, kuko GPS ntihora ikora; ntibikwa kuri seriveri. | P1 |
| **F-03** | Aho umukiriya ari habikwa kuri telefoni ye gusa; ashobora kuhahindura cyangwa kuhasiba igihe cyose. | P1 |
| **F-04** | Bikoreshwa: gutondeka 'Hafi yanjye', kwerekana ku iduka niba riri mu karere ke, no gushyirwa mu butumwa bwa WhatsApp. | P1 |
| **F-05** | Urubuga **ntiruzereka igihe cyo gutanga cyangwa igiciro cya delivery**. Umucuruzi ashobora kwandika inyandiko ngufi y'iduka ('Gutanga i Kigali mu masaha 2'). | P1 |

## G. Ibisa n'ibyarebwe

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **G-01** | 'Wari warabirebye': ibicuruzwa 20 bya nyuma, bibikwa kuri telefoni (`localStorage`). | P1 |
| **G-02** | 'Ibisa na cyo' ku rupapuro rw'igicuruzwa (amategeko muri 9.2). | P1 |
| **G-03** | 'Bishya' na 'Bikunzwe cyane' ku rupapuro rw'imbere (amategeko muri 9.3). | P1 |
| **G-04** | 'Ushobora gukunda': ibicuruzwa by'icyiciro umukiriya akunze kureba cyane, bibarwa kuri telefoni ye. | P2 |

## H. Notification n'amakuru

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **H-01** | Gusaba kwakira notification: icyifuzo cyacu bwite (si icya mushakisha) kigaragara nyuma y'uko umukiriya asuye ≥2 cyangwa agashyira igicuruzwa mu gare. Iyo yanze, ntikigaruka mu minsi 30. | P3 |
| **H-02** | Web push (OneSignal) ifite tags nibura: akarere, icyiciro gikunzwe, size y'inkweto (OneSignal Free igira imipaka y'ama-tags; reba 8.4). | P3 |

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **H-03** | Ubwoko: 'Bishya' (inshuro 1 mu cyumweru, ibishya byiza ku bakurikiye icyiciro cyabo); 'Igiciro cyagabanutse' ku byo yasuye (P4). | P3 |
| **H-04** | Imipaka: ntibirenze 3 mu cyumweru kuri umuntu umwe; nta kohereza hagati ya 21:00 na 07:00 (CAT). | P3 |
| **H-05** | Urupapuro 'Imiterere ya notification' rwo kuzimya ubwoko bumwe cyangwa bwose, igihe cyose. | P3 |
| **H-06** | Imeri (si itegeko): umukiriya ashobora kuyandika ngo yakire amakuru; ububiko bw'imeri bufite itariki n'inyandiko y'uko yabyemeye, n'umurongo wo kwikuraho muri buri imeri. Kohereza biciye muri serivisi y'ubuntu (urugero Brevo; imipaka igenzurwa mbere yo gutangira). | P3 |
| **H-07** | Urubuga **ntiruzakusanya nimero za WhatsApp z'abakiriya** ngo ruzohereze ubutumwa rusange. Amakuru ya WhatsApp yoherezwa binyuze muri WhatsApp Channel y'urubuga. | P3 |
| **H-08** | Admin ashobora kohereza ubutumwa rusange (broadcast) rimwe mu cyumweru. | P3 |

## I. Konti n'ibicungwa by'umucuruzi

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **I-01** | Kwinjira: nimero ya telefoni (+250…) n'ijambo ry'ibanga (nibura inyuguti/imibare 6). Mu buryo bwa tekiniki nimero ihindurwa imeri y'imbere (urugero `250788123456@abacuruzi.invalid`) kugira ngo dukoreshe Supabase Auth nta kwishyura SMS. | P1 |
| **I-02** | Gusubiza ijambo ry'ibanga: nyir'urubuga arisubiza abisabwe (kuri WhatsApp). Kuri SMS/OTP: P3. | P1 |
| **I-03** | Urupapuro rw'imbere rw'umucuruzi: ibicuruzwa biriho, byarebwe (iminsi 7), wa clicks (iminsi 7), n'ibyo gukosora ('5 bishaje: Iracyahari?'). | P1 |
| **I-04** | Urutonde rw'ibicuruzwa bye: gushakisha, ibyungurura (Biriho/Byashize/Byahishwe), utubuto 'Cyashize', 'Kopora', 'Hindura igiciro'. | P1 |
| **I-05** | Umwirondoro w'iduka: izina, slug, logo, ifoto y'umutwe, ibisobanuro (≤200), WhatsApp, telefoni, akarere/umurenge, akarango, amasaha, inyandiko yo 'Gutanga' (≤120). | P1 |
| **I-06** | Akabuto 'Gerageza link ya WhatsApp yanjye' kugira ngo umucuruzi yemeze ko nimero ikora. | P1 |
| **I-07** | Imipaka ishyirwaho na admin: ibicuruzwa 150 biriho, amafoto 5 kuri buri gicuruzwa (bihindurwa). | P1 |
| **I-08** | Kubika ibyanditswe (draft) mu bubiko bwa mushakisha (`IndexedDB`) iyo internet ivuyeho, bikoherezwa nigarutse. | P2 |
| **I-09** | Kuvugurura igiciro cy'icyiciro cyose ukoresheje ijanisha (urugero 'gabanya 10% ku nkweto zose'). | P3 |
| **I-10** | Kohereza ibicuruzwa biturutse muri WhatsApp (ifoto + izina → igicuruzwa). Bisaba amafaranga y'integuza; ntibitegerejwe vuba. | P4 |

## J. Kugenzura umucuruzi

Amabwiriza yose ari mu gice cya 10.1. Ibisabwa bya tekiniki:

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **J-01** | Ifishi y'ibyangombwa: amazina, nimero y'indangamuntu, ifoto y'indangamuntu (imbere n'inyuma), selfie ifashe indangamuntu, telefoni, nimero ya WhatsApp, ifoto y'aho acururiza, ibyo acuruza, TIN (niba ayifite). | P2 |
| **J-02** | Ibyangombwa bibikwa mu bubiko **bwihariye** (private), bufunguwe na admin gusa, kandi buri gufungura byandikwa mu `audit_log`. | P1 |
| **J-03** | Imiterere y'umucuruzi: `pending` → `approved` / `needs_changes` / `rejected` / `suspended`. Ibicuruzwa bigaragara gusa iyo ari `approved`. | P1 |

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **J-04** | Umucuruzi yemera amategeko n'amasezerano mu nyandiko, hakabikwa itariki na verisiyo y'inyandiko yemeye. | P1 |
| **J-05** | Mu cyiciro cya 1 abacuruzi 3 bakora ubugenzuzi bumwe n'abandi (ariko nyir'urubuga ni we ubwemeza), kugira ngo tugerageze uburyo. | P1 |

## K. Admin panel

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **K-01** | Urupapuro rw'imbere: abategereje kwemezwa, ibirego bishya, ibicuruzwa bishya (24h), imibare y'urubuga. | P1 |
| **K-02** | Gufungura ibyangombwa by'umucuruzi (ifoto nini, ihuza n'iyo yohereje) no kwemeza/gusubiza inyuma/kwanga ushyiraho impamvu. Ubutumwa bwateguwe (templates) bwo kumusubiza. | P1 |
| **K-03** | Guhagarika no gusubizaho umucuruzi; guhisha igicuruzwa icyo ari cyo cyose; bibikwa mu `audit_log`. | P1 |
| **K-04** | Urutonde rw'ibicuruzwa byose rufite gushakisha no kungurura. | P1 |
| **K-05** | Ibirego: urutonde, imiterere (gishya/gikemurwa/cyakemuwe), icyakozwe. | P1 |
| **K-06** | Gucunga ibyiciro, uturere, amagambo ahwanye (synonyms), amabara, sizes, imipaka y'ibicuruzwa. | P2 |
| **K-07** | Kwinjira kwa admin bisaba kwemeza mu byiciro bibiri (2FA, TOTP). | P2 |
| **K-08** | Gukuramo amakuru (CSV) y'abacuruzi n'ibicuruzwa. | P2 |
| **K-09** | Kohereza broadcast ya notification. | P3 |
| **K-10** | Abafasha (moderators) bafite uburenganzira buto (kuzimya ibicuruzwa, gukemura ibirego, nta kureba indangamuntu). | P3 |

## L. Gutanga ikibazo (report)

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **L-01** | Buto kuri buri gicuruzwa no ku iduka: 'Tanga ikibazo'. Ubwoko: Impimbano, Igiciro cy'ikinyoma, Ifoto itari iy'ukuri, Umucuruzi yariganyije, Ibibujijwe, Ibindi. | P1 |
| **L-02** | Ibisobanuro (≤500 inyuguti), nimero y'utanga ikibazo (si itegeko, yo gusubizwa). | P1 |
| **L-03** | Kwirinda ibyo gukoreshwa nabi: Cloudflare Turnstile (ubuntu) n'umupaka w'ibirego 5 ku munsi kuri telefoni imwe. | P1 |
| **L-04** | Igicuruzwa kibonye ibirego ≥3 byo mu masaha 24 gihita gihishwa by'agateganyo, admin agasuzuma. | P2 |

## M. Imibare

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **M-01** | Ibikorwa byanditse: `product_view`, `wa_click`, `call_click`, `cart_add`, `share_click`. Byose bibarwa buri munsi kuri buri gicuruzwa n'iduka (reba 7.2). **Nta bikorwa byanditse kimwe kimwe**, nta IP, nta nimero z'abakiriya. | P1 |
| **M-02** | Gukumira kwiyongera kw'imibare: ikintu kimwe kibarwa rimwe kuri session kuri umunsi (session ID ibitswe kuri telefoni). | P1 |
| **M-03** | Cloudflare Web Analytics (ubuntu, nta cookies) ku bwoko rusange bw'abasura. | P1 |

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **M-04** | Dashboard y'umucuruzi: iminsi 7 na 30; ibicuruzwa 5 byarebwe cyane; igicuruzwa gikunzwe kuri WhatsApp. | P2 |

## N. Amatangazo n'ubukode (nyuma)

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **N-01** | Imitwe 2 yabikiwe amatangazo ku mpapuro zimwe (hagati y'imirongo y'imbere, mu bisubizo by'ishakisha), igaragara gusa iyo `ADS_ENABLED=true`. Isanzwe izimye. | P4 |
| **N-02** | Amatangazo ntagomba gutinza paji: apakirwa nyuma y'ibindi byose, ntabe hejuru y'ibicuruzwa, kandi ntabe ku rupapuro rw'igicuruzwa mu mwaka wa mbere. | P4 |
| **N-03** | Imbonerahamwe y'iduka ifite `plan` (isanzwe 'free') na `plan_expires_at` (null) kugira ngo ubukode bwongerwemo nta guhindura database. | P1 |
| **N-04** | Gucunga ubukode: kwandika ibyishyuwe, kwibutsa no guhagarika igihe cyarangiye. | P4 |

## O. Ibindi

| Nimero | Ibisabwa | Icyiciro |
|---|---|---|
| **O-01** | Impapuro zihamye: Ibyerekeye, Amategeko n'amabwiriza, Politiki y'ibanga, Amabwiriza y'abacuruzi, Ubufasha, 404, 'Nta internet'. | P1 |
| **O-02** | `sitemap.xml` na `robots.txt` bikorwa ubwabyo; ibicuruzwa byemewe gusa bikajya muri sitemap. | P2 |
| **O-03** | Ubufasha: buto ya WhatsApp y'urubuga (nimero ya nyir'urubuga) n'amasaha asubizwa. | P1 |
| **O-04** | Inyandiko ya 'cookie/storage': urubuga ntirukoresha cookies z'amatangazo; rukoresha `localStorage` ku igare n'ibyo umukiriya yakunze. Ibi byanditswe muri Politiki y'ibanga. | P1 |

# 6. Uburyo bwo gushyiraho igicuruzwa

Iyi niyo ntangiriro y'ukuri y'urubuga: **abacuruzi nibabona gushyiraho ibicuruzwa bigoye, urubuga ruzaba ubusa.** Intego ni **amasegonda 60** kuva ku ifoto kugeza ku 'Bika', kuri telefoni, n'ukuboko kumwe. Aha ni ho tugomba kwita cyane kuruta ahandi hose.

## 6.1 Amahame y'ifishi

- **Ifoto mbere.** Amafoto atangira koherezwa nyuma y'uko umucuruzi ayatoranyije, *mu gihe agikomeza kuzuza ifishi* (upload mu ruhande). Ibi ni byo bituma 60 s bishoboka.
- **Buto aho kwandika.** Icyiciro, sizes, amabara n'imiterere bihitwamo nta kwandika. Kwandika ni izina n'igiciro gusa.
- **Imibare yo mu mibare.** Igiciro gikoresha urubuga rw'imibare (numeric keypad) kandi cyerekanwa nka '45 000'.
- **Kwibuka.** Ifishi yibuka icyiciro n'imibare uwo mucuruzi aheruka guhitamo. 'Kopora' ifata igicuruzwa cyose igahindura ifoto n'izina gusa.
- **Nta gutakaza.** Buri ntambwe ibikwa nk'igitabo (draft) iyo ivuyeho; internet ivuyeho nta kintu kiribwa.

## 6.2 Intambwe enye

| Intambwe | Igikubiyemo | Itegeko ry'ingenzi |
|---|---|---|
| **1. Ifoto** | Buto nini 'Fata ifoto' (camera) na 'Hitamo muri galerie'. Amafoto 1–5. Ifoto ya mbere ni yo y'ingenzi (ishobora guhindurwa mu kuyikurura). | Kugabanya ifoto kuri telefoni mbere yo koherezwa (reba 8.6). Gukuraho amakuru y'aho ifoto yafatiwe (EXIF GPS). Upload ihita itangira. |
| **2. Icyo ari cyo** | Icyiciro (buto 4) → ubwoko bwo munsi (chips). Izina (≤60 inyuguti, ryateguwe nk'inama). Igiciro (RWF). 'Ongeraho igiciro cya kera' (ihishe, si itegeko). | Izina rirasabwa; igiciro kiri hagati ya 500 na 5 000 000 RWF. Igiciro cya kera kigomba kuruta icy'ubu. |
| **3. Sizes n'amabara** | Sizes ziri mu bwoko bw'icyiciro (6.4): gukanda izihari. Amabara (chips). Imiterere: Nshya / Mitumba. Ikirango (si itegeko). 'Hasigaye' kuri buri size (si itegeko). | Size 1 nibura. Ikirango: reba 6.3 ku 'Original'. |
| **4. Suzuma** | Isura y'igicuruzwa nk'uko umukiriya azakibona. Buto 'Bika kandi utangaze' na 'Bika nk'igitabo'. | Utangaze bisaba amafoto yose yarangije koherezwa. Nyuma yo kubika: 'Kopora' na 'Sangiza kuri WhatsApp'. |

## 6.3 Ibyo bicungwa: imiterere, ikirango, ukuri

- **Imiterere (condition):** 'Nshya' cyangwa 'Mitumba' (yakoreshejwe). Iyi ni ngombwa kuko ari byo bigena igiciro n'icyizere ku isoko ry'u Rwanda.
- **Ikirango (brand):** umucuruzi yandika cyangwa agahitamo (Nike, Adidas, Puma…). Iyo ahisemo ikirango, **ategekwa kwemeza ko igicuruzwa ari 'Original'** (checkbox + inyandiko). Ibicuruzwa bitari original **ntibishobora kwitwa izina ry'ikirango**; bigaragara nka 'Nta kirango' (urugero 'Inkweto za siporo z'abagabo').
- **Impamvu:** kugurisha 'copy' nk'original ni ikinyoma gikomeye kandi gishobora kuzana ibibazo by'umutungo bwite mu by'ubwenge ku rubuga. Ni itegeko ryo mu mabwiriza y'abacuruzi (reba 10.5) kandi buri wese ashobora gutanga ikirego.

## 6.4 Ibyiciro, sizes n'amabara

| Icyiciro | Ubwoko bwo munsi | Sizes |
|---|---|---|
| **Inkweto** (P1) | Z'abagabo · Z'abagore · Z'abana · Siporo (sneakers) · Sandali n'inkweto zoroheje · Inkweto zo mu bwiza (zo gusohokana) · Bote | Abakuru: 36 37 38 39 40 41 42 43 44 45 46 47. Abana: 20–35 (buri size). |
| **Imyenda** (P1) | Abagabo · Abagore · Abana · Imyenda ya siporo · Ankara n'imyenda gakondo · Ibikoti n'amajire (jackets, hoodies) | Inyuguti: XS S M L XL XXL 3XL. Amapantalo: ikibuno 28–42. Abana: imyaka (0–1, 2–3, 4–5, 6–7, 8–10, 11–14). |
| **Amasogisi n'imyenda y'imbere** (P1) | Amasogisi y'abagabo · Y'abagore · Y'abana · Imyenda y'imbere · Imitako n'ibikoresho byo mu rugo | Isogisi: 36–40, 41–45, ubunini bumwe. Imyenda y'imbere: S–XXL. |
| **Ibindi byo kwambara** (P1) | Amasakoshi · Imikandara · Ingofero · Amasaha · Indorerwamo · Imitako · Ibitambaro n'ibisa | Ubunini bumwe (nta size) cyangwa S/M/L ku bishobora. |
| **Ibikoresho by'amatelefoni** (P3) | Ibifata (cases) · Ibirinda ecran · Chargers · Earphones · Power banks · Ibindi | Nta size. Ahubwo: 'Ubwoko bwa telefoni bujyana' (urugero iPhone 13, Samsung A14). |

**Amabara** (urutonde ruhitamo): Umukara, Umweru, Imvi, Ubururu, Umutuku, Icyatsi, Umuhondo, Ikawa, Roza, Amabara menshi. Admin ashobora kongeramo andi (K-06).

## 6.5 Ibisabwa mu mirima y'ifishi

| Umurima | Ubwoko | Itegeko? | Amategeko |
|---|---|---|---|
| photos | ifoto 1–5 | Yego | JPG/PNG/WebP/HEIC; nyuma yo kugabanya buri imwe ≤ 150 KB; ubugari ntarengwa 1000 px |
| category_id | ihitamo | Yego | Ihitamo mu byiciro 5 n'ubwoko bwo munsi |
| name | inyandiko | Yego | 5 kugeza kuri 60 inyuguti; nta nimero ya telefoni, nta link, nta nyuguti nini zose |
| price | umubare wuzuye | Yego | 500 – 5 000 000 RWF |
| old_price | umubare wuzuye | Oya | Iyo ihari igomba kuba iruta 'price'. (P2: igomba guhura n'igiciro cyabayeho muri price_history mu minsi 60.) |
| condition | ihitamo | Yego | new \| used |
| brand | inyandiko/ihitamo | Oya | ≤40 inyuguti; isaba 'authentic' = true |
| authentic | yego/oya | Iyo hari brand | Umucuruzi yemeza ko ari original |
| variants[] | urutonde | Yego | Size 1 nibura; buri size ifite in_stock (yego/oya) na qty (si itegeko, 0–999) |
| colors[] | ihitamo | Oya | Amabara ≤ 5 ku gicuruzwa |
| description | inyandiko | Oya | ≤300 inyuguti; nta link, nta nimero ya telefoni |

## 6.6 Ibikorwa byihuse

- **Kopora:** ikora igicuruzwa gishya gisa na cyo, umucuruzi agahindura ifoto n'izina gusa.
- **Cyashize:** buto imwe ihindura `status` igashyira kuri `sold_out` (ntihindura ifoto). 'Gisubizeho' ikigarura.
- **Hindura igiciro:** akadirishya gato; igiciro cya kera gishyirwamo ubwacyo iyo igiciro kigabanutse (umucuruzi ashobora kukihakana).
- **Iracyahari?:** iyo igicuruzwa kimaze iminsi 30 nta cyakozwe, umucuruzi abona ikarita ifite 'Yego' / 'Cyashize'. Nyuma y'iminsi 45 gihishwa ubwacyo (P2).
- **Imipaka:** ibicuruzwa 150 biriho kuri buri mucuruzi mu ntangiriro (bishobora guhindurwa na admin).

# 7. Amakuru abikwa

Ububiko bw'amakuru ni **PostgreSQL** (muri Supabase). Amazina y'imbonerahamwe n'imirima yanditswe mu Cyongereza (snake_case), kuko ari ko azandikwa muri code. Ibi bishushanyo ni **ihame**; uwubaka ashobora kongeramo imirima ariko ntagomba gukuramo ibyo urubuga rukeneye.

## 7.1 Imbonerahamwe z'ingenzi

| Imbonerahamwe | Icyo ibika |
|---|---|
| **areas** | Uturere n'imirenge y'u Rwanda (reba Ibyongeweho B). |
| **profiles** | Abakoresha bafite konti (abacuruzi, admin, abafasha): uruhare, imiterere yo kwemezwa, itariki yemeye amategeko. |
| **seller_kyc** | Ibyangombwa by'umucuruzi (indangamuntu, selfie, ifoto y'iduka). **Ni amakuru yihariye cyane; admin gusa.** |
| **shops** | Iduka rya buri mucuruzi: izina, slug, WhatsApp, aho rihererereye, imipaka, plan. |
| **categories** | Ibyiciro n'ubwoko bwo munsi, n'ubwoko bwa sizes (size_set). |
| **products** | Ibicuruzwa: izina, igiciro, imiterere, imiterere y'igicuruzwa (status), n'ikoreshwa mu gushakisha. |
| **product_images** | Amafoto ya buri gicuruzwa (inzira muri R2, ubugari, ubunini). |
| **product_variants** | Sizes n'amabara bya buri gicuruzwa, n'ibisigaye. |
| **price_history** | Impinduka z'ibiciro (zo kwemeza ibiciro bya kera bidahimbano). |
| **reports** | Ibirego byatanzwe n'abakiriya. |
| **subscribers** | Abemeye notification/imeri, n'icyo bemeye n'igihe. |
| **product_daily_stats, shop_daily_stats** | Ibibarwa buri munsi (views, wa_clicks...). Nta makuru bwite. |
| **synonyms** | Amagambo ahwanye yo gushakisha (urugero nayiki → nike). |
| **settings** | Imipaka n'ibimenyetso (feature flags): `ADS_ENABLED`, imipaka y'ibicuruzwa, verisiyo y'amategeko. |
| **audit_log** | Ibikorwa byose bya admin n'ifungurwa ry'ibyangombwa. |

## 7.2 Imirima ya buri mbonerahamwe

### profiles

| Umurima | Ubwoko | Icyo ari cyo |
|---|---|---|
| id | uuid, PK | Ni id ya Supabase Auth (auth.users.id) |
| role | text | seller \| admin \| moderator |
| full_name | text | Amazina y'umucuruzi (ahagaragara ni izina ry'iduka) |
| phone, whatsapp | text | Imibare mu buryo mpuzamahanga, nta +: 2507XXXXXXXX |
| status | text | pending \| approved \| needs_changes \| rejected \| suspended |
| status_reason | text | Impamvu y'iyo miterere (yerekanwa umucuruzi) |
| approved_by, approved_at | uuid, timestamptz | Ni nde wemeje n'igihe |
| terms_version, terms_accepted_at | text, timestamptz | Verisiyo y'amategeko yemeye n'itariki |

| Umurima | Ubwoko | Icyo ari cyo |
|---|---|---|
| created_at | timestamptz | Itariki y'ifungurwa |

### seller_kyc (RLS: admin n'umucuruzi ubwe gusa; ibikubiyemo bibikwa neza)

| Umurima | Ubwoko | Icyo ari cyo |
|---|---|---|
| seller_id | uuid, PK | → profiles.id |
| full_name_on_id | text | Amazina nk'uko ari ku ndangamuntu |
| national_id_number | text | Nimero y'indangamuntu (bishobora gufungwa muri Supabase Vault, P2) |
| id_front_path, id_back_path, selfie_path, shop_photo_path | text | Inzira z'amadosiye muri bucket yihariye 'kyc' |
| tin | text, null | TIN niba ayifite |
| business_type | text | Urugero: iduka, ucuruza mu rugo, ucuruza kuri interineti |
| phone_verified_at, whatsapp_verified_at | timestamptz | Igihe admin yahamagaye akemeza nimero |
| submitted_at, reviewed_at, reviewed_by, review_notes | … | Ubugenzuzi |

### shops

| Umurima | Ubwoko | Icyo ari cyo |
|---|---|---|
| id | uuid, PK | |
| owner_id | uuid | → profiles.id (mu cyiciro cya 1 buri mucuruzi afite iduka 1) |
| slug | text, unique | Inyuguti nto, imibare n'utuyobora; igera kuri 30 |
| name | text | Izina ry'iduka (≤40) |
| logo_path, cover_path | text, null | Amafoto muri R2 |
| about | text | ≤200 inyuguti |
| whatsapp, phone | text | Nimero zo kuvugana |
| district_id, sector_id | int → areas | Aho rihererereye |
| landmark, hours_text, delivery_text | text | Akarango, amasaha, inyandiko yo 'Gutanga' (≤120) |
| product_limit | int | Isanzwe 150 |
| plan, plan_expires_at | text, timestamptz | Isanzwe 'free', null (ubukode bw'ejo hazaza) |
| created_at | timestamptz | |

### products

| Umurima | Ubwoko | Icyo ari cyo |
|---|---|---|
| id | uuid, PK | |
| shop_id, category_id | uuid, int | → shops, → categories |
| name, description | text | ≤60 na ≤300 |
| price, old_price | int | RWF (imibare yuzuye). old_price null iyo nta kigabanyijwe |
| condition | text | new \| used |

| Umurima | Ubwoko | Icyo ari cyo |
|---|---|---|
| brand, authentic | text, bool | Ikirango n'icyemezo cy'uko ari original |
| colors | text[] | Amabara |
| status | text | draft \| active \| sold_out \| hidden \| expired |
| hidden_reason | text, null | Impamvu yo guhisha (iyo admin ahishe) |
| search | tsvector | Yakozwe ubwayo mu izina, ikirango, icyiciro; igenewe gushakisha (9.1) |
| created_at, updated_at, last_confirmed_at, sold_out_at | timestamptz | Itariki; last_confirmed_at ni iy'ubugenzuzi 'Iracyahari?' |

### Izindi mbonerahamwe

| Imbonerahamwe | Imirima |
|---|---|
| **categories** | id, parent_id, slug, name_rw, name_en (null), sort, size_set, active, phase |
| **product_images** | id, product_id, position, path_full, path_thumb, width, height, bytes |
| **product_variants** | id, product_id, size_label, color (null), in_stock (bool), qty (int, null) |
| **price_history** | id, product_id, price, changed_at |
| **reports** | id, target_type (product\|shop), target_id, type, details, reporter_phone (null), session_hash, status (new\|in_progress\|resolved), resolution, created_at, resolved_by, resolved_at |
| **subscribers** | id, kind (push\|email), external_id (OneSignal id cyangwa imeri), district_id, top_category_id, consent_at, consent_version, unsubscribed_at |
| **product_daily_stats** | product_id, day (date), views, wa_clicks, call_clicks, cart_adds, shares (PK: product_id + day) |
| **shop_daily_stats** | shop_id, day, views, wa_clicks, call_clicks, shares (PK: shop_id + day) |
| **synonyms** | term, canonical |
| **settings** | key (PK), value (jsonb) |
| **audit_log** | id, actor_id, action, target_type, target_id, details (jsonb), created_at |
| **areas** | id, province, district, sector (null ku karere), parent_id, active |

## 7.3 Amategeko y'uburenganzira muri database (RLS)

RLS ni yo **shingiro** y'umutekano. Amabwiriza y'umukiriya ntahabwa 'service key'; akoresha gusa urufunguzo rusanzwe (anon key), kandi database ikanga ibitemewe.

| Imbonerahamwe | Umukiriya (anon) | Umucuruzi | Admin |
|---|---|---|---|
| products, product_images, product_variants | Gusoma gusa iyo product.status ∈ {active, sold_out} **kandi** umucuruzi ari 'approved' | Gusoma, kwandika, guhindura, gusiba iby'iduka rye gusa | Byose |
| shops | Gusoma iby'abacuruzi 'approved' gusa | Guhindura irye gusa | Byose |
| profiles | Nta na kimwe (ibigaragara bishyirwa muri view) | Gusoma/guhindura ibye (ntahindura role/status) | Byose |
| seller_kyc | **Nta na kimwe** | Kwandika no gusoma ibye (ntahindura reviewed_*) | Gusoma byose; guhindura reviewed_* |
| reports | Kwandika gusa (insert) binyuze muri RPC | Nta na kimwe | Byose |

| Imbonerahamwe | Umukiriya (anon) | Umucuruzi | Admin |
|---|---|---|---|
| subscribers | Kwandika/guhagarika ibye binyuze muri RPC | Nta na kimwe | Gusoma |
| *_daily_stats | Nta kwandika kutaziguye; gukoresha RPC `track_event` gusa | Gusoma iby'iduka rye | Byose |
| audit_log | Nta na kimwe | Nta na kimwe | Gusoma; kwandika bikorwa na seriveri gusa |

## 7.4 Ingero za SQL

```
-- Gushakisha: kubika ijambo rishakwa (tsvector) no kuvugurura ubwayo
alter table products add column search tsvector
  generated always as (
    setweight(to_tsvector('simple', coalesce(name,'')), 'A') ||
    setweight(to_tsvector('simple', coalesce(brand,'')), 'B') ||
    setweight(to_tsvector('simple', coalesce(description,'')), 'C')
  ) stored;
create index products_search_idx on products using gin (search);
create extension if not exists pg_trgm;   -- kwihanganira amakosa y'inyandiko
create index products_name_trgm on products using gin (name gin_trgm_ops);

-- Ibigaragara kuri buri wese: gusa ibyemejwe
create policy "public read products" on products for select
using (
  status in ('active','sold_out')
  and exists (select 1 from profiles p join shops s on s.owner_id = p.id
              where s.id = products.shop_id and p.status = 'approved')
);

-- Umucuruzi ahindura ibye gusa
create policy "seller manages own products" on products for all
using (shop_id in (select id from shops where owner_id = auth.uid()))
with check (shop_id in (select id from shops where owner_id = auth.uid()));

-- Kubara ibikorwa nta makuru bwite (security definer)
create function track_event(p_type text, p_product uuid, p_shop uuid)
returns void language plpgsql security definer as $$
begin
  if p_type not in ('view','wa_click','call_click','cart_add','share') then return; end if;
  insert into product_daily_stats(product_id, day, views, wa_clicks, call_clicks, cart_adds, shares)
  values (p_product, current_date,
          (p_type='view')::int, (p_type='wa_click')::int, (p_type='call_click')::int,
          (p_type='cart_add')::int, (p_type='share')::int)
  on conflict (product_id, day) do update set
    views = product_daily_stats.views + (p_type='view')::int,
    wa_clicks = product_daily_stats.wa_clicks + (p_type='wa_click')::int,
    call_clicks = product_daily_stats.call_clicks + (p_type='call_click')::int,
    cart_adds = product_daily_stats.cart_adds + (p_type='cart_add')::int,
    shares = product_daily_stats.shares + (p_type='share')::int;
end $$;
```

## 7.5 Aho amadosiye abikwa

| Ububiko | Ibirimo | Uburyo bwo kuhagera |
|---|---|---|
| **Cloudflare R2** (bucket 'public', iri kuri domain yawe) | Amafoto y'ibicuruzwa n'amaduka: `products/{shop_id}/{product_id}/{n}_full.webp` na `..._thumb.webp`; `shops/{shop_id}/logo.webp` | Gusoma: buri wese. Kwandika: gusa binyuze muri `/api/upload` igenzura umucuruzi (token ya Supabase) igatanga umukono umara iminota 5. |

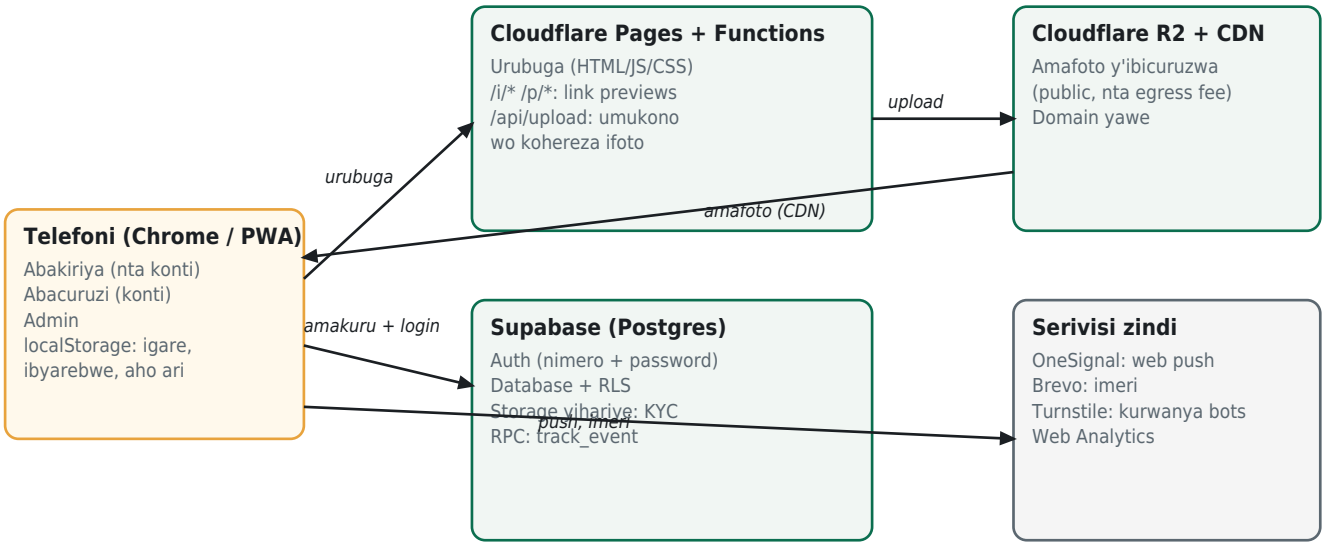
| Ububiko | Ibirimo | Uburyo bwo kuhagera |
|---|---|---|
| **Supabase Storage** (bucket 'kyc', yihariye) | Ibyangombwa by'abacuruzi: `kyc/{seller_id}/{id_front\|id_back\|selfie\|shop}.webp` | Kwandika: umucuruzi ubwe. Gusoma: admin gusa, binyuze muri 'signed URL' imara amasegonda 60. |

## 7.6 Igihe amakuru amara

- **Igicuruzwa cyashize:** kigaragara iminsi 14, hanyuma `hidden`. Cyahishwe iminsi 90: amafoto asibwa muri R2 kugira ngo umwanya udashira.
- **Igicuruzwa kitakozweho iminsi 45:** `expired` (P2).
- **Imibare ya buri munsi:** ibikwa imyaka 2, hanyuma igahurizwa hamwe ku kwezi.
- **Ibyangombwa (KYC):** bibikwa igihe konti ikora; nyuma y'ifungwa rya konti bibikwa igihe cyemejwe n'umunyamategeko (icyifuzo cy'ibanze: amezi 12) hanyuma bigasibwa. Reba igice 11.
- **Subscribers:** uwikuyeho yirengagizwa ako kanya kandi amakuru ye asibwa mu minsi 30.

# 8. Ubwubatsi bwa tekiniki

## 8.1 Incamake




*Ishusho 1: ibice by'urubuga n'uko bihuza. Telefoni zivugana n'urubuga rwa Cloudflare, n'amakuru ya Supabase; amafoto aza ahandi (R2) kugira ngo adatinda.*

## 8.2 Ibikoresho byatoranyijwe

| Igice | Igikoresho | Impamvu |
|---|---|---|
| Urubuga (frontend) | Vite + Preact + TypeScript; CSS yoroheje (nta framework ya UI iremereye); gusimbura impapuro (router) byoroheje | Preact ni nto cyane (~4 KB), ihuye na telefoni zidafite ubushobozi bwinshi; irahagije kuri paji zacu. |
| Kwakira urubuga (hosting) | Cloudflare Pages | Ubuntu, bandwidth itagira imipaka ku madosiye ahamye, CDN ku isi yose. |
| Imirimo ya seriveri | Cloudflare Pages Functions | Gusa ibintu bike: link previews, kuvugana na R2 (umukono), kwemeza token. |
| Database + Auth | Supabase (PostgreSQL) | Auth, database, RLS na Storage byose bifatanye; ubuntu ku ntangiriro. |
| Ububiko bw'amafoto | Cloudflare R2 | 10 GB z'ubuntu kandi nta mafaranga yo gusohora amafoto (egress). |
| Notification | OneSignal (web push) | Web push ni ubuntu; tuyikoresha binyuze muri PWA. |
| Imeri | Brevo (cyangwa nka yo) | Gahunda y'ubuntu; imipaka igenzurwa mbere yo gutangira (P3). |
| Kurwanya bots | Cloudflare Turnstile | Ubuntu, nta CAPTCHA ibabaza. |
| Imibare rusange | Cloudflare Web Analytics | Ubuntu, nta cookies. |
| Code, kopi, kugumisha | GitHub + GitHub Actions | Code mu bubiko bwihariye; gahunda zikora ubwazo. |

| Igice | Igikoresho | Impamvu |
|---|---|---|
| Kureba ko urubuga rukora | UptimeRobot (ubuntu) cyangwa nka yo | Itumenyesha ku imeri/WhatsApp iyo urubuga rubuze. |

## 8.3 Imiterere y'amadosiye (repository)

```
/                     (repository imwe, yihariye)
  src/
    pages/            home, search, product, shop, cart, seller/*, admin/*, static/*
    components/       ProductCard, SizePicker, FilterSheet, LocationSheet, ...
    lib/              supabase.ts, cart.ts, recents.ts, whatsapp.ts, image.ts, track.ts
    i18n/rw.ts        amagambo yose y'ahagaragara (Kinyarwanda)
  functions/
    i/[slug].ts       link preview y'iduka (OG tags)
    p/[id].ts         link preview y'igicuruzwa
    api/upload.ts     umukono wo kohereza ifoto kuri R2
  supabase/
    migrations/       SQL zose zikurikiranye (imbonerahamwe, RLS, functions)
    seed/             areas.sql, categories.sql
  public/             manifest.webmanifest, sw.js, icons/
  .github/workflows/  deploy.yml, backup.yml, keepalive.yml
  README.md           uko gutangira, impinduka z'ibidukikije (env vars)
```

## 8.4 Imipaka y'ubuntu (yagenzuwe ku wa 13/11/2026)

Ibi byavuye ku mbuga z'abatanga serivisi n'inyandiko zisobanura ibiciro byabo (reba Ibyongeweho D). **Imipaka iracyahinduka**; mbere yo gutangira kubaka, uwubaka agomba kongera kuyisuzuma ku mbuga zabo.

| Serivisi | Imipaka y'ubuntu | Icyo bisobanura ku rubuga rwacu |
|---|---|---|
| Cloudflare Pages | Bandwidth n'ibyakirwa bihamye bitagira imipaka; 500 builds/ukwezi; 1 build icyarimwe; ibikorwa bya Functions bisangira imipaka ya Workers: **100 000 ku munsi**. | Urubuga ubwarwo ntiruzigera rugarukira ku mubare w'abasura. Functions zikoreshwa gusa ahantu hacye (previews, upload), bityo 100 000 ku munsi birahagije mu ntangiriro. Ntitwishingikirize kuri 'server-side rendering' ya buri paji. |
| Cloudflare R2 | 10 GB z'ububiko; ibikorwa 1 miliyoni (kwandika) na 10 miliyoni (gusoma) ku kwezi; **nta egress fee**. | Amafoto 3 kuri igicuruzwa (hafi 375 KB hamwe n'ako kato) ≈ **ibicuruzwa 25 000** mbere yo kurenga 10 GB. Ibi bihagije ku bacuruzi benshi. |
| Supabase (Free) | Database 500 MB; file storage 1 GB; egress 5 GB; 50 000 MAU; **imishinga 2**; **irahagarara nyuma y'icyumweru 1 nta gikoreshwa**; nta kopi za buri munsi. | 1 GB y'amafoto ni nke, ni yo mpamvu amafoto ajya muri R2. Ibicuruzwa 25 000 bifata hafi 100 MB muri database (8.5). Gahunda yo kugumisha no gukora kopi ni ngombwa (8.10). |
| OneSignal (Free) | Web push: abafatabuguzi bageze ku 10 000 kuri buri kohereza; imipaka y'ama-tags 6; ibyo kuri mobile app byagabanyijwe ku 1 000 MAU kuva ku wa 01/10/2026. | Dukoresha web push (PWA), si mobile app, bityo imipaka ya mobile ntitugwa. Tags zacu zitagomba kurenga 6. |
| Cloudflare Turnstile, Web Analytics | Ubuntu. | Nta gikorwa. |

### Igihe cyo kuva ku buntu

| Ikimenyetso | Icyo gukora |
|---|---|
| Database igeze kuri 400 MB, cyangwa urubuga rufite abakoresha bakora buri munsi kandi gupfa rutunguranye bidashoboka | Gukura Supabase kuri gahunda ya Pro (hafi 25 USD ku kwezi, ibiciro byagenzuwe); ikuraho 'pause', yongeraho kopi za buri munsi. |

| Ikimenyetso | Icyo gukora |
|---|---|
| R2 igeze kuri 8 GB | Kuyongera ni 0.015 USD ku GB ku kwezi; gusiba amafoto y'ibicuruzwa byahishwe; kongera imyizerere y'ifoto (WebP ntoya). |
| Functions zigeze kuri 70 000 ku munsi | Gukora previews zikorerwa kuri 'cache', cyangwa kugura gahunda ya Workers ($5 ku kwezi). |
| Gukoresha amatangazo cyangwa ubukode | Ayo mafaranga ni yo yishyura izi serivisi (reba igice 16). |

## 8.5 Ubushobozi

| Ikibazo | Igisubizo cy'ibarwa |
|---|---|
| Ibicuruzwa bingahe mu 10 GB za R2? | Buri gicuruzwa: amafoto 3 × (hafi 120 KB y'ifoto nini + 30 KB y'iyoroheje) ≈ 450 KB; 10 000 000 KB ÷ 450 ≈ 22 000. Gufata 25 000 ni icyizere cyiza gifite kugabanya ifoto. |
| Abacuruzi bangahe? | Abacuruzi 150 × ibicuruzwa 150 = ibicuruzwa 22 500: byuzuza 10 GB. Ni yo mpamvu 'imipaka 150' n'amafoto 5 byashyizweho. |
| Database ni nini gute? | Igicuruzwa kimwe n'amafoto n'sizes ≈ 4 KB muri database; 25 000 × 4 KB = 100 MB. Ni yo mpamvu imibare ibikwa ihurijwe buri munsi aho kuba ibikorwa byose. |
| Abasura bangahe ku munsi? | Urubuga ruhamye (HTML/JS) ntirugira umupaka. Ariko amakuru ava muri Supabase ahura n'**egress ya 5 GB ku kwezi**, ishobora kuba umupaka wa mbere. Ku makuru ≈ 15 KB ku rupapuro (urutonde rw'ibicuruzwa 20 nta mafoto), 5 GB ≈ impapuro 330 000 ku kwezi. Igisubizo: amakuru rusange (ibyiciro, uturere, imirongo y'imbere) abikwa muri cache ya Cloudflare, kandi ibisubizo bigatanga imirima ikenewe gusa. |

## 8.6 Uburyo bw'amafoto

Amafoto ni yo ahenze kuruta ibindi (data y'umukiriya, umwanya w'ubuntu, igihe cy'umucuruzi). Dukora ibi byose **kuri telefoni y'umucuruzi** mbere yo koherezwa:

1. Gusoma ifoto (JPEG/PNG/HEIC) no gukosora uko yafashwe (orientation).
2. **Gukuraho amakuru yose y'ifoto (EXIF)**, cyane cyane aho yafatiwe (GPS), kugira ngo umucuruzi atamenyekana aho atuye. Gukora ifoto nshya kuri `canvas` bibikora ubwabyo.
3. Gukora kopi 2: **ntoya** (ubugari bwa 400 px, WebP ~30 KB) na **nini** (ubugari bwa 1000 px, WebP ~100–120 KB, 'quality' ≈ 0.78). Aho WebP itakora, JPEG.
4. Kohereza zombi kuri R2 binyuze mu mukono w'umunota 5 uvuye kuri `/api/upload` (igenzura ko uwohereza ari umucuruzi wemejwe kandi ko ari mu mipaka ye).
5. Mu rubuga: ifoto ntoya mu mirongo no mu gushakisha; nini ku rupapuro rw'igicuruzwa; `loading='lazy'`; ubugari n'uburebure bwanditswe kugira ngo paji itazamuka igihe amafoto apakirwa (CLS).
6. Nta 'upload' y'ifoto irenze 500 KB nyuma yo kugabanya (itangwa n'urubuga ruyanga).

## 8.7 Umuvuduko, PWA na offline

| Igipimo | Intego |
|---|---|
| JS yose ya mbere (gzip) | Munsi ya 100 KB |
| Paji ya mbere yose (ifite amafoto ya mbere) | Munsi ya 500 KB |
| LCP (ifoto nini igaragara) | Munsi y'amasegonda 2.5 kuri 4G itinze (Lighthouse mobile) |
| CLS (paji yisimbuza) | Munsi ya 0.1 |
| Lighthouse mobile: Performance / Accessibility / SEO / PWA | ≥ 85 / ≥ 90 / ≥ 90 / yujuje |

- **PWA:** `manifest.webmanifest` (izina, amabara, ibimenyetso 192 na 512 px), `service worker` (ububiko bw'ibice by'urubuga; amafoto aheruka kurebwa; paji ya 'Nta internet').
- **Kuvugurura:** verisiyo nshya y'urubuga ikururwa mu bwiherero kandi ikerekana 'Verisiyo nshya irahari. Kanda ngo uyigereho'.
- **Cache:** ibyiciro, uturere n'amazina y'amaduka biba muri cache iminota 10; ibiciro n'ibisigaye ntibishyirwa muri cache (cyangwa gusa isegonda 30).
- **Data nke:** Ntitwakira fonts zo hanze. Dukoresha fonts z'ibikoresho (system fonts) cyangwa imwe nto yashyizwe mu rubuga.

## 8.8 Link previews na SEO

Umucuruzi ashyira link ku status ya WhatsApp. Iyo link igaragara **ifite ifoto, izina n'igiciro**, abantu bayikanda cyane kuruta link yambaye ubusa. Ibi ni ngombwa mu kuzana abasura ba mbere.

- Umuhanda `/p/{id}` (igicuruzwa) na `/i/{slug}` (iduka) ukorwa na Cloudflare Function ikora HTML ntoya ifite `og:title`, `og:description`, `og:image` (ifoto nini) n'igiciro, ku bashakisha n'imbuga zishaka previews (WhatsApp, Facebook, Googlebot).
- Abantu basanzwe bahabwa urubuga rusanzwe (SPA). Iyi nzira yirinda gukoresha Functions kuri buri paji.
- Impapuro za `JSON-LD` (Product, Offer, Organization) ku rupapuro rw'igicuruzwa na `sitemap.xml` bifasha Google (P2).
- `/shakisha` n'impapuro z'amagare ntizijya muri Google (`noindex`).

## 8.9 Kohereza kuri internet

1. Gukora konti: GitHub, Cloudflare, Supabase, OneSignal, Brevo. Zose zandikwa kuri imeri y'urubuga (si iy'umuntu ku giti cye) kandi 2FA irafungurwa.
2. Ibidukikije bibiri: **production** na **staging**. Mbere yo gushyira impinduka kuri production zigeragerezwa kuri staging (Supabase ya staging ni umushinga wa 2 w'ubuntu).
3. `main` ikoherezwa kuri production ubwayo (Cloudflare Pages); ishami rya `staging` kuri staging; buri 'pull request' ifite preview.
4. Impinduka z'ibidukikije (env): `SUPABASE_URL`, `SUPABASE_ANON_KEY` (rusange), `SUPABASE_SERVICE_KEY` (ibanga, Functions gusa), `R2_ACCOUNT_ID/KEY/SECRET/BUCKET`, `ONESIGNAL_APP_ID`, `TURNSTILE_SITE/SECRET`.
5. **Domain:** mu igerageza ikoreshwa `*.pages.dev`, ariko **mbere yo gufungura kuri rubanda** hagurwa domain (hafi 10–15 USD ku mwaka kuri .com; igiciro cya .rw kigenwa n'ababitanga bemewe, gisuzumwa). Impamvu: R2 ikeneye domain yawe ngo ikore nka CDN yo ku rwego rwa production; adresi ya `r2.dev` igenewe igerageza gusa kandi ifite imipaka; Cloudflare isaba domain yawe ku rubuga rukoreshwa na rubanda.
6. Kwemeza ko urubuga rwakozwe neza (reba igice 14) mbere yo gusangiza link.

## 8.10 Kopi z'ububiko, kugumisha Supabase no gukurikirana

- **Kopi (backup):** Supabase Free nta kopi za buri munsi itanga. Dukora iyacu: `.github/workflows/backup.yml` ikora `pg_dump` buri cyumweru ikayibika muri R2 (bucket yihariye, idahuye n'iy'amafoto), ikabika 8 za nyuma. Kopi igeragezwa kugarurwa (restore) nibura rimwe mu kwezi mu cyiciro cya mbere.
- **Kugumisha:** Supabase Free ihagarara nyuma y'icyumweru nta gikoreshwa. `keepalive.yml` ikora ibisabwa byoroheje buri munsi; ariko iyo urubuga rufite abasura bahoraho ntibikenewe.
- **Gukurikirana:** UptimeRobot ireba `/` buri minota 5; ikohereza WhatsApp/imeri nyir'urubuga iyo rubuze. Amakosa ya JavaScript yandikwa muri konsole ya Cloudflare cyangwa Sentry (ubuntu), atabonetse amakuru bwite.

## 8.11 Umutekano wa tekiniki

- **HTTPS** gusa; `Content-Security-Policy`, `X-Content-Type-Options`, `Referrer-Policy` bishyirwa mu `_headers`.
- **RLS** ku mbonerahamwe zose (7.3). 'Service key' ntigera igera ku mushakisha.
- **Ijambo ry'ibanga:** rihishwa na Supabase Auth (bcrypt); nibura inyuguti 6; kugerageza kunyuranya ni 5 mu munota; kunanirwa inshuro 10 bifunga konti iminota 15.
- **Kohereza amadosiye:** kwemera gusa `image/webp`, `image/jpeg`; ubunini ntarengwa 500 KB; umukono wa R2 ufite izina rya dosiye rizwi ritakwemera 'path traversal'.
- **Gukumira ibisubirwamo:** Turnstile ku mafishi (ikirego, kwiyandikisha); imipaka y'ibikorwa kuri IP/session ku `track_event` n'ibirego.
- **XSS:** inyandiko zose zakozwe n'abacuruzi (izina, ibisobanuro) zirerekanwa nk'inyandiko, ntizigera zishyirwa nka HTML; nta link yemewe mu izina no mu bisobanuro (6.5).
- **Ibyangombwa:** bucket yihariye; 'signed URLs' z'amasegonda 60; buri gufungura kwandikwa muri audit_log; kwinjira kwa admin kusaba 2FA (K-07).
- **Konti:** nyir'urubuga ni we ufite uburenganzira bwose bwa GitHub, Cloudflare na Supabase; abubaka bahabwa uburenganzira bukenewe gusa kandi bukavanwaho umushinga urangiye.

# 9. Uko urubuga 'rutekereza' (algorithm)

Nta bwenge bw'ubukorano (AI) bukenewe. Amategeko yoroshye, asobanutse, ashobora kugeragezwa no guhindurwa, arahagije kandi ntahenda. **Imibare yatanzwe hano (uburemere, imipaka) ni ibyo gutangiriraho**; izahindurwa nyuma yo kureba amakuru nyayo.

## 9.1 Gushakisha

1. **Gutegura ijambo:** kugabanya mu nyuguti nto, gukuraho utuyobora n'ibimenyetso, gukuraho imyanya irenga.
2. **Kwagura ku magambo ahwanye:** buri jambo risuzumwa muri `synonyms` (reba ingero hepfo).
3. **Guhuza:** (a) ijambo ritangira nk'iryanditswe muri `search` (`to_tsquery('simple','nik:*')`); **cyangwa** (b) ubushuti bw'inyuguti (`pg_trgm similarity > 0.3`) ku izina n'ikirango; ibi bihanganira amakosa nka 'nayiki'.
4. **Gutondeka:** `score = 2·ts_rank + similarity + 0.3·(biri mu bubiko) + 0.2·(bishya mu minsi 14) + 0.1·(iduka riri mu karere k'umukiriya).`
5. **Ibyungurura** (icyiciro, size, igiciro, ibara, imiterere, akarere) bikoreshwa nk'ibigomba kuba byose (AND) *mbere* yo gutondeka.
6. **Nta bisubizo:** kwerekana ibisa cyane (urugero ijambo rimwe ryakuweho), n'ibyiciro bifitanye isano.

| Ijambo ryanditswe | Rihindurwa nka | Impamvu |
|---|---|---|
| nayiki, naiki, nike | nike | Amakosa y'inyandiko akunze kuboneka |
| adida, adidasi | adidas | Uko abantu babivuga |
| isogisi, isogise, amasogisi | amasogisi | Ubwinshi n'ubumwe |
| basket, sneakers, bote za siporo | inkweto za siporo | Amazina y'ibisa |
| shati, chemise | ishati | Ururimi rwavanze |
| jean, jeans, ijini | ipantalo ya jean | Imvugo rusange |
| tshirt, t-shirt, tishati | tishati | Imyandikire itandukanye |

Admin yongera cyangwa akosora aya magambo ahwanye (K-06) nta gukora code.

## 9.2 'Ibisa na cyo'

Ku gicuruzwa runaka **X**, ibicuruzwa 8 byatoranywa ku buryo bukurikira (bitarimo X, bitashize, by'amaduka yemejwe). Buri kimwe gihabwa amanota:

| Itegeko | Amanota |
|---|---|
| Ubwoko bwo munsi bumwe n'ubwa X (urugero 'Inkweto za siporo z'abagabo') | +40 |
| Ikirango kimwe | +20 |
| Igiciro kiri hafi: `20 × max(0, 1 − \|p − pX\| / (0.4·pX))` (hagati ya −40% na +40% by'igiciro cya X) | 0 kugeza +20 |
| Iduka riri mu karere kamwe n'ak'umukiriya | +10 |
| Iduka ritandukanye n'irya X (kugira ngo hatagaragara ibya rimwe gusa) | +10 |
| Size umukiriya yahisemo ('Size yanjye') ihari | +5 |
| Bikunzwe: `min(10, popularity_7d / 10)` | 0 kugeza +10 |

Ibisubizo 8 bitangwa, nta iduka rimwe ririmo birenga 3. Iyo nta bihagije mu bwoko bumwe, ubwoko bwo hejuru (icyiciro) bukoreshwa.

## 9.3 'Bishya' na 'Bikunzwe cyane'

- **Bishya:** gutondeka ku `created_at` iramanuka, gusa ibifite ifoto n'iduka ryemejwe. Nta iduka rirenza ibicuruzwa 2 mu mirongo 8 ya mbere.
- **Bikunzwe cyane:** `popularity_7d = views + 2·shares + 3·cart_adds + 5·(wa_clicks + call_clicks)` mu minsi 7 ya nyuma, ukuyemo ibyahishwe/bishize, nta iduka rirenza 2.
- **Kugira ngo ibishya bibone amahirwe:** mu mirongo 'Bikunzwe', umwanya 1 muri 4 uhabwa igicuruzwa cyaje mu masaha 72 ya nyuma.
- **Byagabanyijwe:** ibifite `old_price` n'ijanisha ≥ 10%, bitondetswe ku ijanisha rinini.

## 9.4 'Wari warabirebye' n'ibyo umukiriya akunda (kuri telefoni gusa)

| Ikibikwa muri localStorage | Imiterere | Ikoreshwa |
|---|---|---|
| recents | Urutonde rw'ibicuruzwa 20 bya nyuma rufite igihe | 'Wari warabirebye' |
| interest | Imibare y'icyiciro: +1 kureba, +3 gushyira mu gare, +5 WhatsApp; igabanywa ×0.9 buri minsi 7 | 'Ushobora gukunda', guhitamo icyiciro mu notification |
| cart | Urutonde rw'ibyo mu gare (id, size, ibara, umubare, igiciro cyabonywe) | Igare |
| location | Akarere, umurenge, akarango | Hafi yanjye, ubutumwa bwa WhatsApp |
| mysize | Size y'inkweto n'iy'imyenda | Ibyungurura |
| sid | Session ID y'agateganyo | Kubara view imwe gusa |

**Nta na kimwe muri ibi kiva kuri telefoni.** Urubuga ntirubika amateka y'ibyo umuntu yasuye ku seriveri; ibi ni byo bituma nta ngorane z'amakuru bwite zijyanye na byo mu cyiciro cya 1.

## 9.5 Ibiciro, igabanywa n'ibisigaye

- **Ijanisha:** `round((old_price − price) / old_price × 100)`; rigaragara gusa iyo ≥ 5%.
- **Igiciro cya kera kidahimbano (P2):** `old_price` yemerwa gusa iyo ihwanye n'igiciro cyabayeho muri `price_history` mu minsi 60 ya nyuma. Ibi bikurikiranwa n'urubuga, ntibisabe admin kugenzura.
- **Hasigaye N:** kigaragara iyo `qty` yanditswe kandi 1 ≤ qty ≤ 3. Iyo qty ikubye 0, size ihita yizimya ubwayo.
- **Igicuruzwa cyashize:** iyo size zose zashize cyangwa umucuruzi akanze 'Cyashize', `status = sold_out`.

## 9.6 Kuvugurura ibicuruzwa bishaje

| Iminsi nta kintu cyakozwe | Igikorwa |
|---|---|
| 30 | Umucuruzi abona ikarita 'Iracyahari?' (Yego / Cyashize). 'Yego' ivugurura `last_confirmed_at`. |
| 45 (P2) | Igicuruzwa gihishwa ubwacyo (`expired`); umucuruzi ashobora kugisubizaho n'akanya gato. |
| 14 nyuma yo gushira | Gihisha `hidden`. |
| 90 nyuma yo guhishwa | Amafoto asibwa muri R2. |

## 9.7 Notification yiteguye (P3)

1. Buri wa Gatanu saa 16:00 (CAT; umunsi n'isaha bigeragezwa) hakorwa ubutumwa bumwe kuri buri 'segment' (akarere + icyiciro gikunzwe).

2. Hatoranywa ibicuruzwa 6 bishya by'iminsi 7 ya nyuma mu cyiciro cy'iyo segment, bibanje iby'akarere kayo; iyo bitageze ku 3, ntiwoherezwa.
3. Ubutumwa: ifoto ya mbere, izina, igiciro, link; umutwe: 'Ibishya mu [icyiciro]'.
4. Imipaka iracyubahirizwa: ≤3 mu cyumweru, nta kohereza 21:00–07:00.

# 10. Icyizere, kugenzura n'umutekano w'ubucuruzi

Umuhemu umwe ashobora kwangiza isoko ryose. Nyir'urubuga yavuze ko urubuga atari rwo rwishingira ubucuruzi, ariko umukiriya wariganyijwe akenshi avuga 'urubuga rwanyibye'. Ni yo mpamvu **kumenya abacuruzi**, **gukemura vuba** n'**ubutumwa busobanutse** ari byo shingiro ry'urubuga.

## 10.1 Kwiyandikisha no kugenzura umucuruzi (KYC)

### Amakuru yose asabwa

| Ikisabwa | Impamvu | Icyiciro |
|---|---|---|
| Amazina yose, nimero y'indangamuntu | Kumenya umuntu mu gihe habaye ikibazo | P2 |
| Ifoto y'indangamuntu (imbere n'inyuma) | Guhuza n'amazina n'umwirondoro | P2 |
| Selfie ifashe indangamuntu | Kwemeza ko nyirayo ari we | P2 |
| Nimero ya telefoni n'iya WhatsApp | Kuvugana no kwemeza | P1 |
| Ifoto y'aho acururiza (iduka, ububiko cyangwa aho akorera) | Kwemeza ko hariho koko | P2 |
| Ibyo acuruza n'ubwoko bw'ubucuruzi | Kwemeza ko bijyanye n'icyiciro cyacu | P1 |
| TIN (niba ayifite) n'izina ry'ikigo | Kwongera icyizere (si itegeko) | P2 |
| Kwemera amategeko n'amasezerano (itariki na verisiyo) | Kubaha amategeko | P1 |

### Uko ubugenzuzi bukorwa (nyir'urubuga)

1. Umucuruzi yuzuza ifishi; konti ye iba `pending`. Ibicuruzwa ashyiraho bibikwa ariko ntibigaragara.
2. Admin afungura ibyangombwa, **agereranya isura iri kuri selfie n'iri ku ndangamuntu**, akemeza ko amazina ahuye n'ayo umucuruzi yatanze.
3. Admin **ahamagara nimero ya telefoni** y'umucuruzi (akemeza ko iyo nimero ari iye) kandi akandikira WhatsApp (akemeza ko ihuye); abikwa muri `phone_verified_at`.
4. Admin asuzuma ifoto y'aho acururiza n'ibicuruzwa bya mbere (amafoto ari aye? ibiciro bishyize mu gaciro? nta mpimbano?).
5. Icyemezo: **Emeza** (ibicuruzwa bihita bigaragara), **Subiza inyuma** (ashyiraho icyo gukosora), cyangwa **Anga** (n'impamvu). Intego: subiza mu masaha 24.
6. Ubugenzuzi bukorwa **rimwe**; nyuma yaho umucuruzi ashyiraho ibicuruzwa nta gihari.

> **Ibyo ubugenzuzi bushobora kandi budashobora**
> Kugenzura mu ntoki bigabanya cyane abashaka kuriganya (kuko bazi ko bamenyekanye), ariko **ntibikuraho ibyago**: ibyangombwa by'impimbano birahari, kandi nta buryo bwo kubigenzura mu buryo bwikora ku buntu. Ni yo mpamvu hongerwaho guhamagara, kureba ibicuruzwa bya mbere no gukemura ibirego vuba.

## 10.2 Ibyo umukiriya abona ku mucuruzi

| Ikimenyetso | Igisabwa |
|---|---|
| **Yagenzuwe** (agatabo k'icyatsi) | Konti ni `approved`; telefoni n'ifoto y'aho acururiza byemejwe. |
| **Amaze iminsi N ku rubuga** | Bibarwa ku itariki yemerewe. |
| **Aho ari** | Akarere n'umurenge (nta aderesi yuzuye). |
| **Umubare w'ibicuruzwa biriho** | Bibarwa ubwabyo. |

*'Amanota y'abakiriya' (ratings) ntazibaho mu cyiciro cya 1 kuko bisaba kwemeza ko umuntu yaguze koko; byakwica icyizere iyo bikorewe ku buryo butizewe. Birashoboka nyuma y'uko hari uburyo bwo kwemeza ibyaguzwe.*

## 10.3 Gutanga no gukemura ibirego

| Ikirego | Igikorwa cya mbere (mu masaha 24) | Igikorwa gikurikiyeho |
|---|---|---|
| Igiciro cy'ikinyoma / ifoto itari iy'ukuri | Kwandikira umucuruzi, gusaba gukosora mu masaha 48. | Atagikosoye: guhisha igicuruzwa. |
| Impimbano zigurishwa nk'original | Guhisha igicuruzwa ako kanya; kubaza umucuruzi. | Byemejwe: kuburira; byongeye: guhagarika iduka. |
| Umucuruzi yariganyije (yakiriye amafaranga ntatange ibicuruzwa) | Guhisha iduka ako kanya; guhamagara umucuruzi; kwandikira uwatanze ikirego. | Byemejwe: guhagarika burundu, kubika amakuru, no kuyatanga ku nzego z'umutekano iyo zibisabye mu buryo bwemewe n'amategeko. |
| Ibibujijwe (10.5) | Guhisha igicuruzwa. | Kuburira cyangwa guhagarika bitewe n'uburemere. |
| Ikindi | Gusoma no gusubiza. | — |

- Umucuruzi arumvwa mbere yo guhagarikwa burundu (amasaha 48), keretse habaye uburiganya bugaragara.
- Buri cyemezo cyandikwa muri `audit_log` kigaragaza uwakifatiye n'impamvu.
- Uwatanze ikirego asubizwa (iyo yatanze nimero) kandi abwirwa icyakozwe, nta makuru bwite y'umucuruzi atanzwe uretse ayo itegeko ryemera.

## 10.4 Inama z'umutekano ku bakiriya (zigaragara ku rupapuro rw'igicuruzwa no mu gare)

1. Reba igicuruzwa mbere yo kwishyura, cyane cyane iyo uri i Kigali.
2. Ntukishyure amafaranga menshi mbere ku mucuruzi utazi; shakisha ikimenyetso 'Yagenzuwe'.
3. Emeza izina riri kuri nimero ya MoMo mbere yo kohereza.
4. Iyo ugaragaje ikibazo, koresha buto 'Tanga ikibazo'. Tuzagikurikirana.
5. Urubuga ntirwakira amafaranga yawe kandi ntiruhagarikira kwishyura: ni ibyo wumvikanye n'umucuruzi.

## 10.5 Ibibujijwe

- **Impimbano zigurishwa nk'original**, no gukoresha izina ry'ikirango ku gicuruzwa kitari original (6.3).
- Ibicuruzwa byibwe cyangwa bitemewe n'amategeko.
- **Amafoto atari ay'umucuruzi** (yakuwe ahandi cyangwa ku bandi bacuruzi).
- Ibiciro by'ikinyoma, 'igiciro cya kera' kidahimbano, 'bisigaye 1' bitari ukuri.
- Ibicuruzwa bidafite aho bihuriye n'icyiciro cy'urubuga, n'ibijyanye n'imibonano mpuzabitsina, intwaro, ibiyobyabwenge.
- Kwiyitirira iduka ry'undi, gutanga nimero ya WhatsApp itari iy'umucuruzi, n'ubutumwa bw'ihohotera.
- Gushyira nimero cyangwa link mu izina cyangwa mu bisobanuro by'igicuruzwa (kugira ngo bibe ibyiza ku isakara).

## 10.6 Kuburira ibiciro by'ibitaba bisanzwe (P3)

Iyo igiciro cy'igicuruzwa kiri munsi ya 40% by'igiciro kisanzwe (median) cy'icyiciro cyacyo cyangwa kuruta inshuro 3, gishyirwa ku rutonde rwa admin rwo gusuzumwa (ntigihishwa). Gushaka amafoto amwe ku bacuruzi batandukanye (perceptual hash) ni P4.

# 11. Amategeko n'amakuru bwite

**Icyitonderwa gikomeye**
Iki gice ni incamake y'ibyo twabonye mu nyandiko z'abanyamategeko n'amakuru ya leta bigaragara ku mbuga zizewe (reba Ibyongeweho D). **Si inama y'amategeko.** Mbere yo gufungura urubuga kuri rubanda no gukusanya ibyangombwa by'abacuruzi, **umunyamategeko w'u Rwanda agomba gusoma iki gice agatanga umwanzuro**. Ibi ni byo bintu bya mbere bitari bya tekiniki bigomba gukorwa.

## 11.1 Itegeko ry'amakuru bwite (Itegeko N° 058/2021 ryo ku wa 13/10/2021)

Iri tegeko ryerekeye kurinda amakuru bwite n'ubuzima bwite. Rirakurikizwa kuri buri uwo ari we wese ukusanya amakuru bwite y'abantu baba mu Rwanda. Rigira ibi bikurikira (ibyavuye mu nyandiko zagenzuwe):

| Icyo itegeko risaba | Icyo bisobanuye ku rubuga rwacu |
|---|---|
| **Kwiyandikisha** mu rwego rushinzwe umutekano w'ikoranabuhanga mu gihugu (NCSA) nk'ugenzura amakuru (data controller). Icyemezo gitangwa mu minsi 30 nyuma yo gusaba. | Urubuga rukusanya amakuru bwite (indangamuntu, selfie, telefoni, imeri), bityo rugomba kwiyandikisha mbere yo gutangira kuyakusanya. |
| **Kubika amakuru bwite mu Rwanda**: kuyabika hanze y'u Rwanda byemewe gusa iyo ufite icyemezo cya NCSA kikubwira ko wemerewe (ingingo ya 50). Kuyohereza hanze nabyo bifite amategeko (ingingo ya 48 n'iya 49: amasezerano yanditse n'uwo uyohereza). | **Ikoranabuhanga ryacu ribika hanze y'u Rwanda** (Supabase, Cloudflare, GitHub). Ibi bivuze ko dukeneye uruhushya rwa NCSA, cyangwa tukabika amakuru bwite (cyane cyane KYC) muri seriveri iri mu Rwanda. |
| **Kumenyesha NCSA** mu masaha 48 igihe amakuru yamenetse. | Dukeneye uburyo bwo gutahura iyo amakuru yamenetse n'umuntu uzabimenyesha. Byanditswe muri gahunda y'umutekano (8.11). |
| **Uruhushya (consent)** rwumvikana kandi rutaziguye mbere yo gukusanya. | Ifishi ya KYC, imeri, na notification bifite inyandiko isobanura icyo amakuru azakoreshwa, igihe azamara, n'uburenganzira bwo kuyasiba. |
| **Uburenganzira bw'ufitanye na yo isano**: kureba, gukosora, gusiba, kwanga gutunganywa. | Hari umuntu n'uburyo bworoshye bwo kubisaba (imeri/WhatsApp), kandi bigasubizwa. |
| **Gushyiraho DPO** (ushinzwe kurinda amakuru) no gusuzuma ingaruka (DPIA) ku bikorwa biteje ibyago. | Mu ntangiriro nyir'urubuga ni we DPO; DPIA ngufi ikorwa mbere yo gukusanya ibyangombwa. |
| Ibihano birimo n'igifungo mu gihe cy'amakosa akomeye. | Ni yo mpamvu ibi bitagomba gufatwa nk'ibitoroshye. |

## 11.2 Amahitamo agomba gufatwa mbere ya P2

| Uburyo | Icyiza | Ikibi |
|---|---|---|
| **A.** Kwiyandikisha na NCSA, hanyuma gusaba uruhushya rwo kubika no kohereza amakuru hanze (Supabase, Cloudflare...). | Dukomeza gukoresha ikoranabuhanga ry'ubuntu. | Bisaba igihe n'inyandiko (amasezerano n'abatanga serivisi, ibimenyetso by'umutekano); nta cyemezo cy'uko kizemerwa. |
| **B.** Kwiyandikisha na NCSA, hanyuma kubika **ibyangombwa gusa** (KYC) muri seriveri iri mu Rwanda; ibicuruzwa n'ibindi bikaguma hanze. | Amakuru yihariye cyane ahora mu Rwanda. | Bisaba seriveri yo mu Rwanda (igiciro gito ariko si ubuntu) n'akazi kenshi ko kubaka. |

| Uburyo | Icyiza | Ikibi |
|---|---|---|
| **C.** Mu cyiciro cya 1 nta kwakira amafoto y'ibyangombwa ku rubuga: nyir'urubuga akagenzura abacuruzi 3 mu ntoki (arabazi), akabika inyandiko ku mashini ye yihishe, nta kubishyira mu bicu. | Byihuta, nta ngorane z'amategeko zihutirwa; bihuye n'icyiciro cya 1. | Ntibikomeza kuba byiza igihe abacuruzi benshi batazwi bari kwiyandikisha (P2). |

**Icyifuzo:** gutangira na **C** mu cyiciro cya 1; mu gihe cya P1, gutangira inzira yo kwiyandikisha na NCSA no gusuzuma **A** cyangwa **B** ku nama y'umunyamategeko, ku buryo ibyemezo biboneka mbere y'uko P2 itangira.

## 11.3 Urutonde rw'ibyo gukora (amategeko)

| Igikorwa | Ugikora | Igihe |
|---|---|---|
| Gusaba umunyamategeko w'u Rwanda gusoma iki gice no kuvuga uko tugomba kwiyandikisha | Nyir'urubuga | Mbere y'ifungurwa rya rubanda |
| Kwiyandikisha na NCSA (nk'ugenzura amakuru) | Nyir'urubuga | Mbere yo gukusanya amakuru y'abandi |
| Gusuzuma niba hakenewe uruhushya rwo kubika hanze y'u Rwanda; gufata icyemezo hagati ya A, B, C | Nyir'urubuga + umunyamategeko | Mbere ya P2 |
| Kwiyandikisha nk'ubucuruzi mu rwego rubishinzwe (urugero RDB) no gusuzuma imisoro mbere yo kwinjiza amafaranga (amatangazo, ubukode) | Nyir'urubuga + umucungamari | Mbere ya P4 |
| Kwandika inyandiko zo muri 11.4 no kuzisomesha umunyamategeko | Nyir'urubuga | Mbere y'ifungurwa rya rubanda |
| Gushyiraho gahunda yo kumenyesha amakuru yamenetse (masaha 48) | Nyir'urubuga + uwubaka | Mbere ya P2 |
| Gusuzuma amategeko arengera umuguzi n'ay'ubucuruzi bwo kuri interineti ku nyandiko z'urubuga | Umunyamategeko | Mbere y'ifungurwa rya rubanda |

## 11.4 Inyandiko z'amategeko zigomba kwandikwa

| Inyandiko | Ingingo z'ingenzi |
|---|---|
| **Amategeko n'amabwiriza yo gukoresha urubuga** | Urubuga ni ikibuga gusa; rutaba uruhande mu masezerano y'ubugure; nta kwishyura ku rubuga; uburenganzira bwo guhisha ibirimo no guhagarika konti; ibibujijwe; uko amakimbirane akemurwa. |
| **Politiki y'ibanga** | Amakuru akusanywa (uko akusanywa, impamvu, aho abikwa, igihe amara, uwo ayoherezwa); localStorage; uburenganzira bwa nyir'amakuru; uko wavugana na DPO; kumenyesha amakuru yamenetse. |
| **Amasezerano y'umucuruzi** | Amakuru y'ukuri; kubahiriza ibibujijwe; kubahiriza ibyo yemereye abakiriya; uruhushya rwo kwerekana amafoto ku rubuga; kwemera ko amakuru ye ashobora gutangwa ku nzego z'umutekano; gukemura ibirego; gufungwa kwa konti. |
| **Amabwiriza y'ibibujijwe** | Urutonde muri 10.5. |
| **Gutanga ikirego no gukuraho ibirimo** | Uko ikirego gitangwa, igihe gisubizwamo, uko umucuruzi yiregura, uko ubujurire bukorwa. |

# 12. Isura n'imikoreshereze

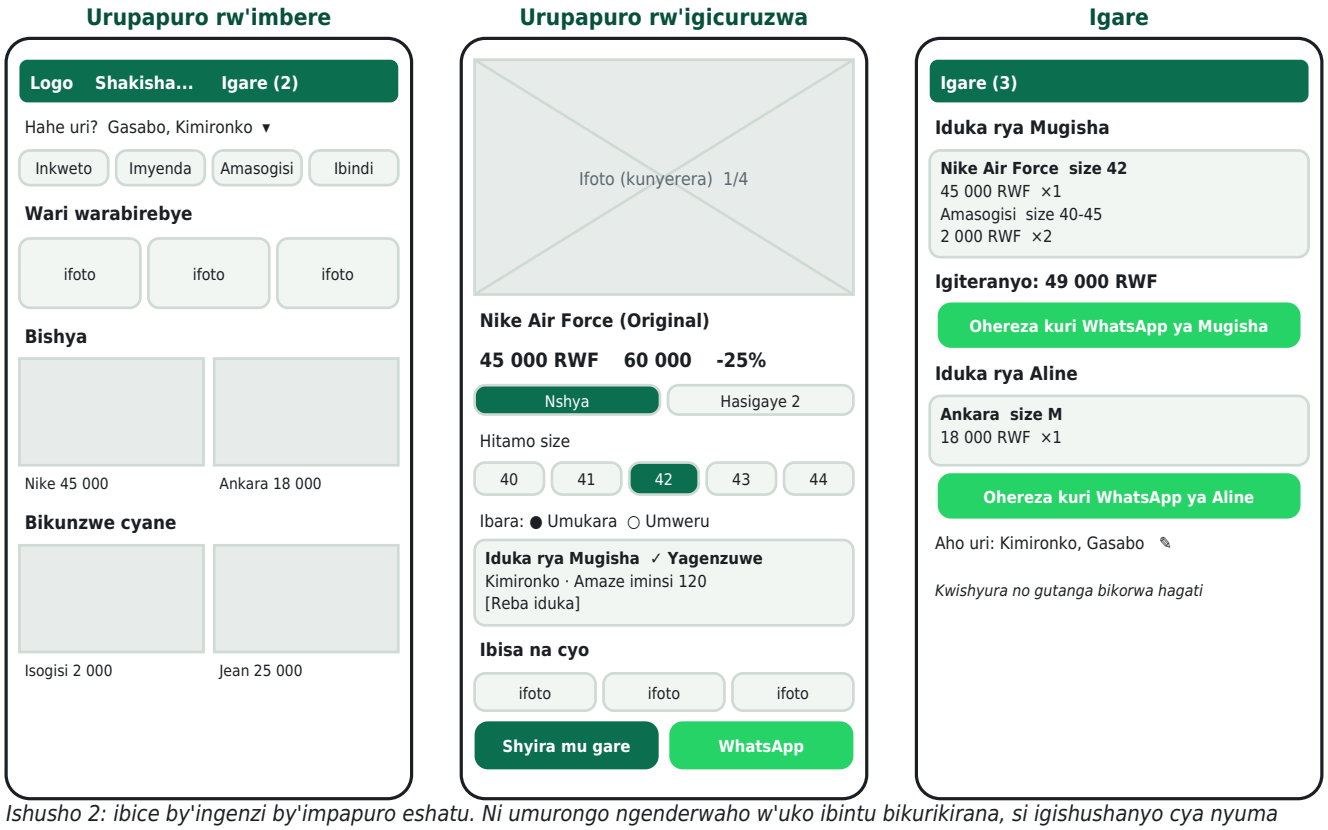
## 12.1 Amahame y'igishushanyo

1. **Igikorwa kimwe nyamukuru kuri buri paji.** Ku rupapuro rw'igicuruzwa: 'Shyira mu gare'. Mu gare: 'Ohereza kuri WhatsApp'. Ku ifishi: 'Bika'.
2. **Ututubuto tunini.** Ahantu hose ho gukanda hagira nibura 44 × 44 px; utubuto twa 'Shyira mu gare' n'ak'WhatsApp turi hasi y'ecran kandi dukomeza kugaragara (sticky).
3. **Igiciro kigaragara cyane** kurusha andi mazina yose ku gicuruzwa. Ifoto ni yo ya mbere, atari inyandiko.
4. **Gusoma byoroshye.** Inyandiko y'ingenzi ≥ 16 px; ubudasa bw'ibara ≥ 4.5:1; nta nyandiko iri ku ifoto idafite ahantu hakwiye kuyisomera.
5. **Amabara make.** Icyatsi nk'ikimenyetso nyamukuru; icyuma/umuhondo nk'ikimenyetso cy'icyitonderwa; icyatsi cya WhatsApp gusa ku butumwa bwa WhatsApp.
6. **Kugenda kworoshye:** nta animation ndende; nta 'popup' ibangamira inyandiko ya mbere; 'skeleton' (aho ifoto izaba) aho gutegereza ecran irimo ubusa.
7. **Byose ku ntoki:** gukoresha urubuga n'ikiganza kimwe kuri telefoni y'iyoroheje.

## 12.2 Ibimenyetso by'igishushanyo (design tokens)

| Ikintu | Agaciro | Ikoreshwa |
|---|---|---|
| Ibara nyamukuru | #0B6E4F (icyatsi) | Ututubuto tw'ingenzi, ibimenyetso, imitwe |
| Ibara ry'icyitonderwa | #E8A33D (umuhondo-orange) | Gushimangira, 'Hasigaye N', ibyagabanyijwe |
| Inyandiko | #1B1F23 | Inyandiko isanzwe |
| Inyuma | #FFFFFF na #F1F6F3 | Paji, ibikoresho |
| Ikosa | #C0392B | Amakosa n'ibyago |
| WhatsApp | #25D366 | Buto za WhatsApp gusa |
| Inyandiko (font) | Inyuguti z'ibikoresho (system-ui, 'Segoe UI', Roboto, 'Noto Sans', sans-serif) | Nta kuyisaba hanze (data) |
| Ingano | Isanzwe 16 px; ibyingenzi 20–22 px; igiciro 20 px bold; utuntu 13 px | |
| Umwanya | Gahunda ya 4 px (4, 8, 12, 16, 24) | |
| Inguni | Amakarita 12 px; ututubuto 12 px; amakipe 999 px | |
| Ifoto y'igicuruzwa | Kare 1:1; imirongo 2 mu murongo kuri telefoni | Imirongo ya 'object-fit: cover' |

## 12.3 Ishusho y'ibice by'ingenzi (wireframes)




*Ishusho 2: ibice by'ingenzi by'impapuro eshatu. Ni umurongo ngenderwaho w'uko ibintu bikurikirana, si igishushanyo cya nyuma cy'amabara n'amashusho; uwishyiraho isura (designer) cyangwa uwubaka arayinoza ariko ntahindura uko ibintu bikurikirana.*

## 12.4 Impapuro n'inzira (routes)

| Inzira (URL) | Urupapuro | Abaruhabwa | Icyiciro |
|---|---|---|---|
| `/` | Urupapuro rw'imbere | Bose | P1 |
| `/shakisha` | Ibisubizo by'ishakisha n'ibyungurura (`?q=&c=&size=...`) | Bose | P1 |
| `/c/{icyiciro}` | Icyiciro (nk'ishakisha rifite icyiciro) | Bose | P1 |
| `/p/{id}` | Igicuruzwa | Bose | P1 |
| `/i/{slug}` | Iduka | Bose | P1 |
| `/igare` | Igare | Bose | P1 |
| `/ubufasha, /amategeko, /ibanga, /ibyerekeye` | Impapuro zihamye | Bose | P1 |
| `/abacuruzi` | Amabwiriza y'abacuruzi + 'Saba gucuruza' | Bose | P2 |
| `/injira` | Kwinjira kw'umucuruzi/admin | Abacuruzi, admin | P1 |
| `/umucuruzi` | Urupapuro rw'imbere rw'umucuruzi | Umucuruzi | P1 |
| `/umucuruzi/ibicuruzwa` | Ibicuruzwa bye | Umucuruzi | P1 |
| `/umucuruzi/ongeraho` | Ifishi y'intambwe 4 | Umucuruzi | P1 |
| `/umucuruzi/iduka` | Umwirondoro w'iduka | Umucuruzi | P1 |
| `/umucuruzi/imibare` | Imibare ye | Umucuruzi | P2 |
| `/admin` | Urupapuro rw'imbere rwa admin | Admin | P1 |

| Inzira (URL) | Urupapuro | Abaruhabwa | Icyiciro |
|---|---|---|---|
| `/admin/abacuruzi`, `/admin/ibicuruzwa`, `/admin/ibirego` | Gucunga | Admin | P1 |
| `/admin/imiterere` | Ibyiciro, uturere, synonyms, imipaka | Admin | P2 |

## 12.5 Amagambo y'ahagaragara (Kinyarwanda)

Aya magambo yose ashyirwa mu `i18n/rw.ts`; nta jambo ryandikwa muri code. Umuntu uvuga Kinyarwanda neza agomba kuyasoma mbere yo gufungura.

| Urufunguzo | Inyandiko | Urufunguzo | Inyandiko |
|---|---|---|---|
| search_placeholder | Shakisha inkweto, imyenda… | add_to_cart | Shyira mu gare |
| cart | Igare | whatsapp_btn | Vugana kuri WhatsApp |
| call | Hamagara | sold_out | Cyashize |
| only_left | Hasigaye {n} | condition_new / used | Nshya / Mitumba |
| verified | Yagenzuwe | member_since | Amaze iminsi {n} ku rubuga |
| choose_size | Hitamo size | choose_location | Hahe uri? |
| send_whatsapp_to | Ohereza kuri WhatsApp ya {iduka} | payment_note | Kwishyura no gutanga bikorwa hagati yawe n'umucuruzi. |
| price_changed | Igiciro cyahindutse | empty_cart | Igare ryawe ririmo ubusa. |
| no_results | Nta cyabonetse. Gerageza ijambo ritandukanye. | report | Tanga ikibazo |
| share | Sangiza | offline | Nta internet. Ibyo wari warabonye biracyahari. |
| login | Injira | phone / password | Nimero ya telefoni / Ijambo ry'ibanga |
| add_product | Ongeraho igicuruzwa | take_photo | Fata ifoto |
| pick_gallery | Hitamo muri galerie | save_publish | Bika kandi utangaze |
| save_draft | Bika nk'igitabo | still_available | Iracyahari? |
| mark_sold | Cyashize | copy_product | Kopora |
| pending_review | Konti yawe itegereje kwemezwa. Tuzaguhamagara. | install_app | Shyira urubuga kuri telefoni yawe |
| notif_ask | Ushaka kumenyeshwa ibishya? | filter / sort | Ungurura / Tondeka |
| recent / new / popular | Wari warabirebye / Bishya / Bikunzwe cyane | similar / discounted | Ibisa na cyo / Byagabanyijwe |

# 13. Gahunda yo kubaka

Twubaka mu byiciro. Buri cyiciro kigerageza ikintu kimwe kitaragerwaho, kandi nticyatangira icyakibanjirije kitarangiye. **Igihe cyatanzwe gishingiye ku mushakashatsi umwe w'inararibonye ukora igihe cyose; ni ugereranya.** Ku mushakashatsi ukora igice cy'igihe, ukubye kabiri.

| Icyiciro | Intego | Abarimo | Igihe (ugereranya) |
|---|---|---|---|
| **P0** Imyiteguro | Gutegura ibyo kubaka bitari code | Nyir'urubuga | Icyumweru 1 |
| **P1** Verisiyo ya mbere (MVP) | Gukora urubuga rukora neza n'abacuruzi 3 | Nyir'urubuga + abacuruzi 2 | Ibyumweru 4 kugeza kuri 6 |
| **P2** Gufungura abacuruzi benshi | Kwakira abacuruzi 10–20 bagenzurwa, i Kigali | + abacuruzi 10–20 | Ibyumweru 3 kugeza kuri 4 |
| **P3** Gukura | Notification, igihugu cyose, ibikoresho by'amatelefoni, Icyongereza | Abacuruzi n'abakiriya benshi | Ibyumweru 4 kugeza kuri 6 |
| **P4** Amafaranga | Amatangazo, ubukode, n'ibindi | Urubuga rumaze kumenyekana | Hashingiwe ku bukure |

## P0: Imyiteguro (nta code)

1. Kwemeza izina ry'urubuga, logo ntoya n'amabara (12.2).
2. Gufungura konti zose (GitHub, Cloudflare, Supabase, OneSignal, Brevo) ku imeri y'urubuga, no gufungura 2FA.
3. Gusaba umunyamategeko w'u Rwanda gusoma igice cya 11, no gutangira kwiyandikisha muri NCSA.
4. Gushaka no kwemeza abacuruzi 2 b'inshuti bazatangirana; kubandika amazina, nimero, WhatsApp, akarere n'ibyo bacuruza.
5. Gufata amafoto y'ibicuruzwa nibura 60 hakurikijwe Ibyongeweho C, no gutegura amazina n'ibiciro byabyo.
6. Kwemeza ibyiciro n'ubwoko bwo munsi (6.4) n'urutonde rw'amabara.
7. Kwandika inyandiko z'amategeko zo kugerageza (11.4) no kuzisomesha.
8. Kugura domain, cyangwa gufata icyemezo cy'igihe kizagurwa.

## P1: Verisiyo ya mbere (MVP) — uko bikurikirana

| # | Igikorwa | Ibisabwa | Gihe cyuzuye iyo… |
|---|---|---|---|
| 1 | Repository, ibidukikije bibiri (staging/production), deploy ikora ubwayo | 8.3, 8.9 | Impinduka ku `main` igaragara kuri production mu minota 5 |
| 2 | Database: migrations, areas, categories, RLS, ibizamini bya RLS | 7, 14.3 | Ibizamini by'umutekano byose byatsinze |
| 3 | Auth y'abacuruzi (nimero + ijambo ry'ibanga), uruhare | I-01, I-02 | Umucuruzi yinjira, atinjira iyo yagenewe nabi |
| 4 | Iduka n'umwirondoro; ifishi y'intambwe 4; amafoto (kugabanya, kohereza kuri R2) | 6, I-05, 8.6 | Umucuruzi ashyiraho igicuruzwa mu ≤ 60 s |
| 5 | Gucunga ibicuruzwa: kopora, cyashize, hindura igiciro | 6.6, I-04 | Byose bikora kuri telefoni |
| 6 | Admin ntoya: kwemeza, guhisha, ibirego | K-01 .. K-05 | Admin yemeza umucuruzi akahisha igicuruzwa |
| 7 | Impapuro z'umukiriya: imbere, ishakisha n'ibyungurura, igicuruzwa, iduka | A, B, C, D | Zikora kuri 360 px |

| # | Igikorwa | Ibisabwa | Gihe cyuzuye iyo… |
|---|---|---|---|
| 8 | Igare, aho ari, ubutumwa bwa WhatsApp | E, F, Ibyongeweho A | Ubutumwa bufungura WhatsApp bwanditse neza |
| 9 | Ibisa na cyo, 'Wari warabirebye', bishya/bikunzwe, imibare (track_event) | G, M, 9 | Imibare igaragara muri dashboard |
| 10 | Link previews, PWA, offline, kunoza umuvuduko | 8.7, 8.8 | Lighthouse intego zagezweho |
| 11 | Impapuro zihamye, ikirego, Turnstile | O, L | Ikirego kigera kuri admin |
| 12 | Kopi, kugumisha, gukurikirana | 8.10 | Kopi yagaruwe neza mu kigeragezo |
| 13 | Ibizamini byuzuye (14), gukosora, abacuruzi 3 bakora ibicuruzwa byabo, gufungura ku nshuti | 14 | Urutonde rwo gufungura rwuzuye |

**Ibisabwa byo kuva kuri P1 ujya kuri P2**
(1) Abacuruzi 3 bakorera ku rubuga nibura ibyumweru 4; (2) nibura 200 wa clicks muri rusange; (3) nta kirego gikomeye kitakemuwe; (4) ibizamini byose bya P1 byatsinze; (5) icyemezo cy'amategeko cyo muri 11.2 cyafashwe kandi cyubahirijwe; (6) kopi y'ububiko yagaruwe neza.

## P2: Gufungura abacuruzi benshi

1. Ifishi yo gusaba gucuruza n'ibyangombwa (J-01), ukurikije icyemezo cya 11.2.
2. Urupapuro rwa admin rwo gusuzuma (K-02), ubutumwa bwateguwe, 2FA ya admin (K-07).
3. Ibiciro bya kera bidahimbano (price_history, 9.5), 'Iracyahari?' n'ihishwa ry'ibishaje (9.6).
4. 'Size yanjye', QR code y'iduka, 'Bigurwa hamwe', kubika ibyanditswe bitarohereje (I-08).
5. JSON-LD, sitemap, kurangiza SEO; kwihisha ikirego cya ≥3 (L-04).
6. Gutumira abacuruzi 10–20 i Kigali (nyir'urubuga abigisha mu ntoki).

## P3: Gukura

1. Notification (web push) n'imeri (H); 'Kurikira iduka'; WhatsApp Channel.
2. Uturere twose 30 n'imirenge yose; icyiciro cy'ibikoresho by'amatelefoni.
3. Icyongereza; kuburira ibiciro bidasanzwe (10.6); guhindura ibiciro mu bwinshi (I-09).
4. Abafasha (moderators) n'uburenganzira bwabo; SMS/OTP yo gusubiza ijambo ry'ibanga (iyo amafaranga ahari).

## P4: Amafaranga

- Amatangazo (N-01, N-02); ubukode bw'abacuruzi (N-04); kohereza ibicuruzwa biturutse muri WhatsApp (I-10); ibindi nyir'urubuga azemeza.

# 14. Ibizamini no kwemeza ko byarangiye

## 14.1 Ibikoresho byo kugerageza

| Igikoresho | Impamvu |
|---|---|
| Telefoni ya Android yoroheje (urugero Tecno/Infinix, RAM 2 GB), Chrome | Ni zo abakiriya benshi bakoresha; igerageza ry'ukuri ry'umuvuduko. |
| Telefoni ya Android yo hagati (Samsung A-series), Chrome | Isanzwe. |
| iPhone (Safari) ushyizwe kuri home screen | Web push kuri iPhone ikora gusa iyo urubuga rwashyizwe kuri home screen; gusuzuma PWA. |
| Computer (Chrome, Firefox) | Admin n'abakoresha computer. |
| Chrome DevTools: 'Slow 4G' + 'CPU 4x slowdown' | Kwigana imiyoboro itinze. |

## 14.2 Ibizamini by'imikorere

| # | Ikizamini | Igisubizo gitegerejwe |
|---|---|---|
| 1 | Umucuruzi ashyiraho igicuruzwa (ifoto, izina, igiciro, sizes) ku telefoni | Bikorwa mu ≤ 60 s; igicuruzwa kigaragara ku rupapuro rw'imbere mu gihe kitarenze isegonda 5 nyuma yo kubika. |
| 2 | Ifoto ya 4 MB yoherejwe | Igabanywa kuri telefoni kugera ≤ 150 KB; nta GPS muri yo (reba EXIF); ntibirenga umupaka wa 500 KB. |
| 3 | Internet ivuyeho hagati yo gushyiraho igicuruzwa | Ibyanditswe ntibitakara; byoherezwa internet nigarutse (P2); mu P1 ubutumwa busobanutse. |
| 4 | Gushakisha 'nayiki' | Hagaragara Nike. |
| 5 | Kungurura ku size 42 + igiciro 20 000–50 000 | Hagaragara gusa ibifite size 42 yo mu bubiko n'igiciro mu mpande; URL irahuye nabyo. |
| 6 | Gushyira mu gare ibicuruzwa by'abacuruzi 2 | Igare rigabanywa mu bice 2, buri kimwe gifite igiteranyo cyacyo. |
| 7 | Gukanda 'Ohereza kuri WhatsApp ya X' | WhatsApp ifungura numero ya X, ubutumwa bufite ibintu bya X gusa, igiteranyo n'aho umukiriya ari. |
| 8 | Igicuruzwa gishyizwe 'Cyashize' n'umucuruzi | Kigaragara 'Cyashize', ntigishyirwa mu gare; cyari mu gare kigaragara 'Cyashize'. |
| 9 | Igiciro gihindutse cyari mu gare | Igare rigaragaza 'Igiciro cyahindutse' n'igishya. |
| 10 | Admin ahisha igicuruzwa | Gihita kitagaragara ku bakiriya; link isubizwa 'Iki gicuruzwa ntikiriho'. |
| 11 | Admin ahagarika umucuruzi | Ibicuruzwa bye byose bihita bihishwa mu isegonda 5. |
| 12 | Gusangiza link y'igicuruzwa kuri WhatsApp | Hagaragara ifoto, izina n'igiciro nk'ibanze. |
| 13 | Gutanga ikirego inshuro 6 mu munsi umwe kuri telefoni imwe | Icya 6 kirangwa (imipaka); ibya mbere 5 bigera kuri admin. |
| 14 | Gufungura urubuga kuri 'Slow 4G' | Paji ya mbere igaragara mu ≤ 3 s; JS ≤ 100 KB gzip. |
| 15 | Gushyira urubuga kuri home screen | Rufungurwa nk'app; 'Nta internet' igaragara iyo nta internet. |

## 14.3 Ibizamini by'umutekano (RLS na seriveri)

| # | Igikorwa kigeragezwa | Igisubizo gitegerejwe |
|---|---|---|
| S1 | Umukiriya (anon) asoma `seller_kyc` | Birangwa; nta murongo na umwe. |
| S2 | Umukiriya asoma ibicuruzwa by'umucuruzi utaremezwa/uhagaritswe | Nta bicuruzwa bigaruka. |
| S3 | Umucuruzi A ahindura cyangwa asiba igicuruzwa cya B | Birangwa. |
| S4 | Umucuruzi ahindura `role` ye cyangwa `status` ye | Birangwa. |
| S5 | Kohereza ifoto ya 5 MB, cyangwa idafite ubwoko bwemewe | Birangwa na /api/upload. |
| S6 | Gukoresha 'upload' nta token y'umucuruzi wemejwe | Birangwa (401/403). |
| S7 | `track_event` inshuro 1000 mu munota 1 | Igabanywa n'umupaka; imibare ntisimbuka. |
| S8 | Gushyira HTML/script mu izina cyangwa ibisobanuro | Yerekanwa nk'inyandiko; nta script ikora. |
| S9 | Gufungura ifoto y'ibyangombwa nta ruhare rwa admin | Birangwa; admin afungura ikandikwa muri audit_log; URL iracika mu masegonda 60. |
| S10 | Admin yinjira nta 2FA (P2) | Birangwa. |

## 14.4 Ibisobanuro bya 'byarangiye' (Definition of Done)

- Ibisabwa byose by'icyiciro bitsinze ibizamini byabyo (14.2 na 14.3).
- Nta kosa rikomeye (rihagarika imikorere cyangwa rigaragaza amakuru y'umutekano) ritarakosorwa.
- Intego z'umuvuduko (8.7) zagezweho kuri telefoni nyayo.
- Amagambo yose y'ahagaragara ni Kinyarwanda, yasomwe n'umuntu uvuga neza.
- Code iri mu bubiko, ifite README ivuga uko itangira, n'impinduka z'ibidukikije zanditswe (nta rufunguzo mu code).
- Kopi y'ububiko yagaruwe neza mu kigeragezo.
- Nyir'urubuga yagerageje buri murimo mu ntambwe ze (umucuruzi, admin, umukiriya) kuri telefoni ye.

## 14.5 Urutonde rwo gufungura kuri rubanda

| Igikorwa | Yakozwe? |
|---|---|
| Domain yaguzwe kandi ikora (HTTPS) n'amafoto ya R2 ari kuri domain yawe | ☐ |
| Inyandiko z'amategeko zasomwe n'umunyamategeko kandi ziri ku rubuga | ☐ |
| Iyandikwa muri NCSA ryakozwe (cyangwa icyemezo cyanditswe cyo kutiyandikisha mbere ya P2) | ☐ |
| Ibicuruzwa nibura 60 biriho, byemejwe, bifite amafoto meza | ☐ |
| Abacuruzi 3 bagerageje gushyiraho no guhindura ibicuruzwa byabo | ☐ |
| Buto za WhatsApp na 'Hamagara' zagerageje kuri nimero ya buri mucuruzi | ☐ |
| Kopi yakozwe kandi igeragejwe kugarurwa | ☐ |
| UptimeRobot ikurikirana urubuga; itumenyesha rigeraho | ☐ |
| Cloudflare Web Analytics ikora | ☐ |
| Nimero y'ubufasha ya WhatsApp iri ku rubuga kandi isubizwa | ☐ |
| Ibizamini byose (14.2, 14.3) byatsinze kuri telefoni nyayo | ☐ |

# 15. Ibyago n'uko byakemurwa

| # | Ikibazo gishobora kubaho | Ingaruka | Igisubizo |
|---|---|---|---|
| 1 | Abacuruzi ntibazakoresha urubuga (bakomeza status gusa) | Rukomeye | Gushyiraho igicuruzwa mu 60 s; link imwe y'iduka n'ifoto ya preview; imibare ibereka abarebye n'abakanze WhatsApp (status ntibitanga); gutangira n'abazwi; kubigisha mu ntoki. |
| 2 | Abakiriya ntibazaza (inkoko n'igi) | Rukomeye | Abacuruzi bazana abakiriya babo ku status zabo (link ifite ifoto); WhatsApp Channel; gutangira ahantu hamwe (Kigali); gukomeza gukorera kuri status buri munsi. |
| 3 | Uburiganya bw'umucuruzi, bukangiza izina ry'urubuga | Rukomeye | KYC, guhamagara, kureba ibicuruzwa bya mbere, ikirego cyoroshye, guhisha vuba, inama z'umutekano (10). |
| 4 | Kudakurikiza itegeko ry'amakuru bwite (kubika hanze, kwiyandikisha) | Rukomeye: ibihano | Igice cya 11: umunyamategeko, NCSA, icyemezo A/B/C mbere ya P2; gukusanya amakuru make. |
| 5 | Imipaka y'ubuntu irenga (Supabase 1 GB / pause / R2 10 GB / Functions 100 000 ku munsi) | Hagati | Amafoto muri R2; kugumisha; gukurikirana buri cyumweru; ibimenyetso byo kuva ku buntu (8.4). |
| 6 | Abatanga serivisi bahindura imipaka cyangwa ibiciro (urugero OneSignal yahinduye mu 2026) | Hagati | Imiterere yimukika (Postgres, S3-compatible R2); kopi z'ububiko; kugenzura imipaka buri gihembwe. |
| 7 | Nyir'urubuga ni umuntu umwe: kunanirwa kwemeza, gusubiza, kugenzura | Hagati | Ubutumwa bwateguwe; SLA ikwiriye (24 h); gukura abacuruzi buhoro; abafasha (P3). |
| 8 | Amafoto mabi, ibiciro bitemewe, ibicuruzwa bishaje | Hagati | Amabwiriza y'amafoto (Ibyongeweho C); 'Iracyahari?'; ihishwa ry'ibishaje; kuburira ibiciro bidasanzwe (P3). |
| 9 | Impimbano n'ibibazo by'umutungo bwite w'ibirango | Hagati | Itegeko rya 'Original' (6.3); ikirego; guhisha vuba. |
| 10 | Bots, spam, gukopera ibicuruzwa byose (scraping) | Hagati | Turnstile, imipaka y'ibikorwa, RLS; nimero z'abacuruzi zigaragara ku bushake (zibwirwa mu masezerano). |
| 11 | Ihiganwa: Facebook Marketplace, WhatsApp Catalog, Jumia n'abandi | Hagati | Guhitamo icyiciro cyihariye (imyambaro n'inkweto), Kinyarwanda, umuvuduko, kugenzura abacuruzi, guha abacuruzi imibare badahabwa ahandi. |
| 12 | Nyir'urubuga atabona amafaranga igihe kirekire | Hagati | Ikoranabuhanga ry'ubuntu; ikiguzi gito (domain); gahunda y'amafaranga isobanutse nyuma (igice 16). |
| 13 | Amakosa ya tekiniki akomeye (data yatakaye, urubuga rwapfuye) | Hagati | Kopi z'ububiko; UptimeRobot; staging; ibizamini mbere ya buri gushyira. |

# 16. Amafaranga

## 16.1 Ikiguzi cyo gutangira

| Ikintu | Ikiguzi | Icyitonderwa |
|---|---|---|
| Hosting, database, amafoto, notification, imibare | 0 USD | Mu mipaka y'ubuntu (8.4). |
| Domain | hafi 10–15 USD ku mwaka (.com) | Ikenewe mbere yo gufungura kuri rubanda. Igiciro cya .rw kigomba gusuzumwa. |
| Inama y'umunyamategeko | Ntikizwi: gusuzuma | Igice cya 11. |
| Kwiyandikisha na NCSA | Ntikizwi: gusuzuma | Igiciro n'uburyo bigomba kubazwa NCSA. |
| Kwiyandikisha k'ubucuruzi, imisoro | Ntikizwi: gusuzuma | Mbere y'uko winjiza amafaranga (P4). |
| SMS/OTP yo gusubiza ijambo ry'ibanga | Ikiguzi kuri SMS | P3, iyo amafaranga abonetse. |
| Umwanya n'akazi ko kubaka | — | Ni ukwa nyir'urubuga cyangwa uwo azakoresha; ntibibarwa hano. |

## 16.2 Igihe cyo kuva ku buntu

Reba 8.4. Mu magambo make: **Supabase Pro** (hafi 25 USD ku kwezi) niyo ikunze kuza mbere, iyo database igeze hafi 400 MB cyangwa urubuga rukeneye kudapfa; hakurikiraho R2 (0.015 USD ku GB ku kwezi hejuru ya 10 GB) na Workers (5 USD ku kwezi) iyo previews zigeze hafi ya 100 000 ku munsi.

## 16.3 Inkomoko y'amafaranga mu gihe kizaza

Ibi ni **icyemezo cya nyir'urubuga**. Nta na kimwe kigomba kubakwa mbere ya P4, keretse imbonerahamwe y'iduka ifite `plan` n'`plan_expires_at` (N-03).

| Inkomoko | Igihe | Ibyitonderwa |
|---|---|---|
| **Amatangazo** (Google AdSense cyangwa asa na yo) | Nyuma y'uko urubuga rumaze kumenyekana (imyaka 3–5, nk'uko nyir'urubuga yabigennye) | Amatangazo yishyura make ku musuye muri iki gihugu, bityo akenera abasura benshi. Atinza paji, bityo ashyirwa nyuma kandi ntabe ku rupapuro rw'igicuruzwa mu mwaka wa mbere (N-02). Gusuzuma ko AdSense yemerera u Rwanda kandi uko amafaranga yishyurwa, n'ibisabwa byo kwemerwa. |
| **Ubukode bw'ukwezi bw'abacuruzi** | Nyuma y'imyaka myinshi, abanyarwanda benshi bamaze kumenya urubuga | Abacuruzi ba mbere (abamaze igihe) bashobora guhabwa igihe cy'ubuntu cyangwa igiciro gito, kugira ngo batagenda. Igiciro kizemezwa nyuma yo kuvugana na bo. |
| **Komisiyo** | **Ntizabaho** (icyemezo cya nyir'urubuga) | Urubuga ntirwinjira mu bucuruzi bw'abandi. |

# 17. Ibibazo bisigaye bigomba kwemezwa

Ibi ni ibyemezo bitaraboneka bikenewe mbere cyangwa mu gihe cyo kubaka. Buri kimwe gifite icyifuzo cyacu; nyir'urubuga ahitamo.

| # | Icyemezo | Icyifuzo | Mbere ya |
|---|---|---|---|
| 1 | Izina ry'urubuga, domain na logo | Izina rigufi, ryoroshye kuvuga no kwandika mu Kinyarwanda; domain ihuye n'izina | P0 |
| 2 | Ni nde uzubaka: nyir'urubuga cyangwa uwo azakoresha? | Umushakashatsi umwe w'inararibonye (full-stack); iyi nyandiko niyo ahabwa | P0 |
| 3 | Amakuru bwite: uburyo A, B cyangwa C (11.2) | C muri P1; gufata icyemezo cya A/B mbere ya P2 | P2 |
| 4 | Abacuruzi 2 bazatangira n'ibyo bacuruza | Bamenyekanishwe mu nyandiko mbere yo gutangira | P0 |
| 5 | Umupaka w'ibicuruzwa kuri buri mucuruzi | 150 biriho, amafoto 5 kuri buri gicuruzwa | P1 |
| 6 | Nimero n'amasaha by'ubufasha | Nimero imwe ya WhatsApp, buri munsi 08:00–20:00 | P1 |
| 7 | Gusoma amagambo y'urubuga (Kinyarwanda) | Umuntu uvuga neza asoma 12.5 n'inyandiko z'amategeko | P1 |
| 8 | Icyongereza ku rubuga | P3 | P3 |
| 9 | Ibyiciro bya nyuma (6.4) | Gutangirana n'ibyo bine; amatelefoni muri P3 | P0 |
| 10 | Umunyamategeko uzasoma igice cya 11 | Uzi amategeko y'ikoranabuhanga n'amakuru bwite | P0 |
| 11 | Serivisi y'imeri (H-06) | Brevo cyangwa yindi, nyuma yo kugenzura imipaka y'ubuntu | P3 |
| 12 | Itariki yo gufungura kuri rubanda | Nyuma y'urutonde rwo muri 14.5 | P1 |

## Ibyo nyir'urubuga yamaze kwemeza (ntibivuguruzwa)

- Urubuga ni ikibuga; nta komisiyo; nta kwishyura ku rubuga; abacuruzi n'abakiriya bavugana ubwabo.
- Abacuruzi bazagenzurwa (indangamuntu, telefoni, ifoto y'aho bacururiza, ibyo bacuruza).
- Gutangirana n'abacuruzi 3 (nyir'urubuga n'inshuti 2 zizewe).
- Urubuga rworoshye, rwihuta, rukoresha ibikoresho by'ubuntu.
- Amafaranga azaza nyuma: amatangazo, hanyuma ubukode bw'ukwezi.

# Ibyongeweho A: Ubutumwa bwa WhatsApp

Ubutumwa bwakozwe n'urubuga bushyirwa muri link ya `https://wa.me/{nimero}?text={ubutumwa}`, ubutumwa bukandikwa mu buryo bwa `encodeURIComponent`. {nimero} ni iy'umucuruzi mu buryo mpuzamahanga nta '+' (urugero `250788123456`).

## A.1 Igicuruzwa kimwe

```
Muraho {izina_ry_iduka},
nabonye iki gicuruzwa ku [URUBUGA]:

{izina_ry_igicuruzwa}
Size: {size}  Ibara: {ibara}
Igiciro: {igiciro} RWF
{link_y_igicuruzwa}

Mbese kirahari?
Aho ndi: {akarere}, {umurenge} {akarango}
{link_y_ikarita_niba_ihari}
```

## A.2 Igare (ubutumwa bumwe kuri buri mucuruzi)

```
Muraho {izina_ry_iduka},
nashakaga ibi bicuruzwa nabonye ku [URUBUGA]:

1. {izina_1} - size {size_1}, {ibara_1} - {igiciro_1} RWF x {umubare_1}
2. {izina_2} - size {size_2}, {ibara_2} - {igiciro_2} RWF x {umubare_2}

Igiteranyo: {igiteranyo} RWF
(Igiciro cya delivery ntikirimo; twumvikane.)

Aho ndi: {akarere}, {umurenge} {akarango}
{link_y_ikarita_niba_ihari}

Mbese birahari? Twumvikane ku kwishyura no kubigeza aho ndi.
```

## A.3 Amategeko yo kubaka ubutumwa

- **Ubunini:** ubutumwa ntibugomba kurenga inyuguti 1 500 (URL ndende ntizihagurukira). Iyo burenze, link z'ibicuruzwa zikurwamo uretse iya mbere, hanyuma imirongo ikagabanywa.
- **Imibare:** ibiciro byanditse nka '45 000' (umwanya hagati y'imibare itatu); ibiteranyo bibarwa n'urubuga ariko umucuruzi ni we wemeza.
- **Kwemeza nimero:** nimero y'umucuruzi igomba guhuza na `^2507[2389]\d{7}$` (MTN na Airtel); izindi ntizemerwa mu ifishi.
- **Aho umukiriya ari:** iyo adahisemo aho ari, umurongo wa 'Aho ndi' ukurwamo, ntukorwe ubusa.
- **Link y'ikarita:** `https://maps.google.com/?q={lat},{lng}`; ishyirwamo gusa iyo umukiriya yakanze 'Ohereza aho ndi'.
- **Gufungura:** kuri telefoni `wa.me` ifungura WhatsApp; kuri computer ifungura WhatsApp Web. Iyo bidakora hagaragara 'Koporora ubutumwa'.
- **Igare rirenga ibintu 10 ku mucuruzi:** ritegekwa kugabanywa mu butumwa bubiri (E-08 igabanya ku 10).

# Ibyongeweho B: Uturere n'imirenge

Aya makuru ashyirwa muri `areas`. Imirenge ya Kigali ikoreshwa mu cyiciro cya 1; imirenge y'utundi turere ishyirwa muri P3. Uwubaka agomba kongera kugenzura urutonde n'inyandikire y'ubuyobozi mbere yo kuyishyira muri database.

## B.1 Intara n'uturere 30

| Intara | Uturere |
|---|---|
| **Umujyi wa Kigali (3)** | Gasabo, Kicukiro, Nyarugenge |
| **Amajyaruguru (5)** | Burera, Gakenke, Gicumbi, Musanze, Rulindo |
| **Amajyepfo (8)** | Gisagara, Huye, Kamonyi, Muhanga, Nyamagabe, Nyanza, Nyaruguru, Ruhango |
| **Iburasirazuba (7)** | Bugesera, Gatsibo, Kayonza, Kirehe, Ngoma, Nyagatare, Rwamagana |
| **Iburengerazuba (7)** | Karongi, Ngororero, Nyabihu, Nyamasheke, Rubavu, Rusizi, Rutsiro |

## B.2 Imirenge y'Umujyi wa Kigali (35)

| Akarere | Imirenge |
|---|---|
| **Gasabo (15)** | Bumbogo, Gatsata, Gikomero, Gisozi, Jabana, Jali, Kacyiru, Kimihurura, Kimironko, Kinyinya, Ndera, Nduba, Remera, Rusororo, Rutunga |
| **Kicukiro (10)** | Gahanga, Gatenga, Gikondo, Kagarama, Kanombe, Kicukiro, Kigarama, Masaka, Niboye, Nyarugunga |
| **Nyarugenge (10)** | Gitega, Kanyinya, Kigali, Kimisagara, Mageragere, Muhima, Nyakabanda, Nyamirambo, Nyarugenge, Rwezamenyo |

# Ibyongeweho C: Amabwiriza y'amafoto n'abacuruzi ba mbere

## C.1 Uko gufata ifoto nziza (ni yo igurisha)

1. **Urumuri rw'umunsi.** Hafi y'idirishya cyangwa hanze, nta zuba rikomeye ryagwa ku gicuruzwa cyonyine. Ntukoreshe flash.
2. **Inyuma hasa kandi hera.** Urukuta rwera, ikarito yera cyangwa umwenda woroshye. Nta bintu byinshi inyuma.
3. **Igicuruzwa cyuzuye mu ifoto** (cyane cyane ifoto ya mbere); ntugicike.
4. **Telefoni ihagaze** kandi ifoto ihuza n'icyo ushaka kwerekana (kare cyangwa yegeranye).
5. **Amafoto 3 kugeza kuri 5:** imbere, ku ruhande, inyuma, hasi (inkweto) cyangwa imbere y'umwenda n'ikimenyetso (label) cyawo.
6. **Isuku.** Siba umukungugu ku nkweto n'amasakoshi; kunyunyuza imyenda.
7. **Nta watermark yuzuye ifoto, nta nimero ya telefoni mu ifoto.** Izina ry'iduka rigaragara ku rubuga.
8. **Amafoto ni ayawe.** Ntukoreshe amafoto yakuwe ahandi cyangwa ku bandi bacuruzi.

## C.2 Imyiteguro y'abacuruzi ba mbere (P0)

| Icyo gutegura | Ibisobanuro |
|---|---|
| **Amazina y'ibicuruzwa** | Izina rigufi risobanutse (urugero 'Inkweto za siporo z'abagabo - umukara'). Ikirango gusa iyo ari original. |
| **Ibiciro** | Ibiciro by'ubu byanditswe nk'imibare. Igiciro cya kera gusa niba kidahimbano. |
| **Sizes** | Urutonde rw'izihari kuri buri gicuruzwa, n'umubare igihe uzi neza. |
| **Imiterere** | Nshya cyangwa mitumba, kuri buri kimwe. |
| **Amafoto** | Ya 3–5 kuri buri gicuruzwa, nibura ibicuruzwa 20 kuri buri mucuruzi. |
| **Umwirondoro w'iduka** | Izina, logo (si itegeko), nimero ya WhatsApp n'iya telefoni, akarere n'umurenge, akarango, amasaha, inyandiko yo 'Gutanga'. |
| **Ibyangombwa** | Indangamuntu, telefoni, ifoto y'aho acururiza (nk'uko icyemezo cy'amakuru bwite cya 11.2 kibyemerera). |

# Ibyongeweho D: Aho amakuru yavuye

Ibyo imipaka, ibiciro n'amategeko byagenzuwe ku wa **13 Ugushyingo 2026** hifashishijwe izi nyandiko. Zisobanura ibyo twemeje; **ntizisimbura** kongera gusuzuma ku mbuga z'abatanga serivisi mbere yo gutangira, kuko ibiciro n'imipaka bihinduka.

| Ingingo | Inkomoko | Icyo yemeje |
|---|---|---|
| Supabase Free | cloudzero.com/blog/supabase-pricing; makerkit.dev/blog/saas/supabase-pricing; activepieces.com/blog/supabase-pricing-free-tier-limits-pro-costs-egress (yo ku wa 30/09/2026) | Database 500 MB, file storage 1 GB, egress 5 GB, 50 000 MAU, imishinga 2, pause nyuma y'icyumweru 1; Pro ni 25 USD ku kwezi. |
| Cloudflare R2 | developers.cloudflare.com/r2/pricing | 10 GB-ukwezi, ibikorwa 1M (A) na 10M (B), egress ni ubuntu; 0.015 USD ku GB hejuru. |
| Cloudflare Pages/Workers | developers.cloudflare.com/pages/functions/pricing; developers.cloudflare.com/workers/platform/pricing; temps.sh/blog/cloudflare-pages-free-tier-limits-2026 | Pages: 500 builds/ukwezi, ibihamye bitagira imipaka; Functions zisangira 100 000 ku munsi. |
| OneSignal | documentation.onesignal.com/docs/en/billing-faq; hackceleration.com (OneSignal alternatives) | Mobile push na in-app bigabanyijwe ku 1 000 MAU guhera 01/10/2026; web push na imeri bikomeza; web push ≤ 10 000 abafatabuguzi kuri buri kohereza; imipaka y'ama-tags 6. |
| Itegeko ry'amakuru bwite | dlapiperdataprotection.com (Rwanda); pwc.com/rw (Data protection in Rwanda; Navigating cross-border data transfer); dti.eui.eu (ingingo ya 50); hlc.com | Itegeko N° 058/2021; kwiyandikisha na NCSA mu minsi 30; kubika mu Rwanda keretse hari icyemezo (ingingo ya 50); amasaha 48 yo kumenyesha; ibihano. |

### Ibyo tutagenzuye (bigomba kugenzurwa mbere yo kubifatiraho umwanzuro)

- Imipaka n'ibiciro bya Brevo (cyangwa indi serivisi y'imeri).
- Igiciro cya domain ya .rw n'iya .com ku batanga bo mu Rwanda.
- Ibisabwa n'ibiciro byo kwiyandikisha muri NCSA n'icyemezo cyo kubika hanze y'u Rwanda.
- Niba AdSense yemerera u Rwanda n'uko amafaranga yishyurwa.
- Ibiciro by'SMS/OTP mu Rwanda.
- Urutonde rwa nyuma rw'uturere n'imirenge n'imyandikire yabyo mu rubuga rw'ubuyobozi.

### Verisiyo y'inyandiko

| Verisiyo | Itariki | Ibyahindutse |
|---|---|---|
| 1.0 | 13 Ugushyingo 2026 | Inyandiko ya mbere yuzuye. |